

PARIKSHA365 — STUDY NOTES

Environment & Ecology — Common Book

SSC, RRB, State PSC — Biomes, Climate, Acts,
Protected Areas

SSC

RRB

STATE PSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

Examiner's Blueprint — What Environment Questions Are Really About

EXAM ALERT

Environment & Ecology is one of the most rewarding GA topics — examiners repeat the same themes year after year.

Here's how they think:

- **1. Conventions and treaties with years** — "Ramsar Convention year?", "Montreal Protocol?", "COP-28?" Every SSC CGL paper.
- **2. Protected areas — "first" and "only" facts** — India's first NP, only floating NP, only Asiatic lion habitat, latest tiger reserve. Every paper.
- **3. National symbols + conservation projects** — National animal, aquatic animal, heritage animal; Project Tiger vs Elephant vs Cheetah. 1–2 marks every paper.
- **4. Laws + Acts with years** — EPA 1986, WLPA 1972, Forest Rights Act 2006, NGT Act 2010. Year is almost always part of the question.
- **5. India's climate commitments** — Net-zero by 2070, 50% non-fossil by 2030, PANCHAMRIT (5 points from Glasgow 2021).

Everything else — biogeochemical cycles, IUCN categories, biodiversity hotspots, pollution parameters — appears occasionally. Covered here in depth; revise in the order above.

Priority order for your study time:

1. International conventions table (Part Y) — year + key fact for each
2. India's acts with years (Part F, Part X) — especially EPA 1986, WLPA 1972, NGT 2010
3. Conservation projects — Tiger, Elephant, Cheetah, Dolphin, Lion (Part Z)
4. Key protected areas — first NP, floating NP, Gir, Kaziranga, Sundarbans (Part C)
5. National symbols (Part Z)
6. Climate commitments — PANCHAMRIT, Paris Agreement, net-zero 2070 (Part E)
7. Biodiversity hotspots (Part B)
8. Pollution metrics — AQI, BOD, DO (Part D)
9. Biogeochemical cycles (Part A)
10. Ecological pyramids and energy flow (Part A)

Current counts to know (Verified June 2026):

Item	Value (June 2026)
Tiger Reserves	58 (56th = Guru Ghasidas - Tamor Pingla, CG; 57th = Ratapani, MP; 58th = Madhav, MP)
Biosphere Reserves	18 in India (12 listed under UNESCO MAB)
Ramsar Sites	100 (Milestone reached June 5, 2026 with Jai Prakash Narayan Bird Sanctuary, UP)
Tiger population	~3,167 (2022 census); 2026 census ongoing

Item	Value (June 2026)
National Parks	106
Wildlife Sanctuaries	~570

How to read this book — your real study load

Total pages: 40. Don't let the page count scare you.

Mandatory reading — pages 1 to 31 (78% of the book)

That's the full syllabus. Master those 31 pages and you've covered every concept an examiner can fairly ask.

Bonus material — pages 32 to 40 (22% of the book)

Reference tables, compilations, and self-test material — useful in your last week of revision OR if you're aiming for a top rank.

Two study tracks

- **Just want to pass?** Read pages 1–31. Skim the appendices once before the exam.
 - **Want to top your batch?** Read the appendices too — they're how 70% becomes 90%.
-

Index — Table of Contents (clickable)

Pages	Part / Chapter	Topic / What it covers
p5–8	PART A	ECOLOGY FUNDAMENTALS
p9–11	PART B	BIODIVERSITY
p12–21	PART C	PROTECTED AREAS AND INDIA'S WILDLIFE
p22–24	PART D	POLLUTION
p25–28	PART E	CLIMATE CHANGE
p29–31	PART F	INDIAN ENVIRONMENTAL LAWS AND INSTITUTIONS
p32–33	PART X	CONSERVATION PROJECTS AND NATIONAL SYMBOLS — <i>Conservation Projects + National Symbols — Tiger, Elephant, Cheetah.</i>
p34–35	PART Y	INTERNATIONAL CONVENTIONS (COMPLETE TABLE) — <i>International Conventions — Ramsar, Montreal, Kyoto, Paris (with years).</i>
p36–40	PART Z	BIODIVERSITY (DETAILED) — <i>Biodiversity Detailed — IUCN categories, hotspots, endemic species.</i>

PART A — ECOLOGY FUNDAMENTALS

Chapter A1 — The Ecosystem

Every ecosystem — pond, forest, ocean — has the same basic structure:

- Things that **make** energy (producers).
 - Things that **consume** it (consumers).
 - Things that **recycle** the leftovers (decomposers).
-

Two components:

- **Abiotic** — non-living: sunlight, temperature, water, air, soil, minerals.
- **Biotic** — living: producers, consumers, decomposers.

Energy Flow in Food Chains

The chain:

Sunlight → **Producers** (green plants, phytoplankton) → **Primary consumers** (herbivores) → **Secondary consumers** (carnivores) → **Tertiary consumers** → **Decomposers** (bacteria, fungi) → nutrient return.

QUICK RECALL

The "90% Tax" Analogy (The 10% Rule): Imagine nature has a strict 90% "handling fee" every time energy changes hands. If Grass makes ₹100 of energy from the sun, the Grasshopper that eats it only gets to keep ₹10. The Frog that eats the grasshopper only gets ₹1. The Snake that eats the frog gets ₹0.10. This is why food chains are short (usually 4 steps max) — the energy simply runs out! It also explains why apex predators (like Tigers) are always rare compared to deer.

KEY FACT

The **10% Rule (Lindeman, 1942)** — only about 10% of the energy at one trophic level is transferred to the next. This is why a vegetarian diet is more energy-efficient and why apex predators require massive hunting territories.

Ecological Pyramids

Pyramid type	Can it be inverted?	Why
Pyramid of numbers	Yes	One tree can support thousands of insects
Pyramid of biomass	Yes	In oceans, phytoplankton biomass can be less than zooplankton at any instant
Pyramid of energy	Never — always upright	Second law of thermodynamics — energy always decreases as it flows up

Ecological Pyramids — Three Types

Energy Pyramid (always upright)

T4 Carnivore T3 Carnivore T2 Herbivore T1 Producer → 10% rule

Numbers Pyramid (can invert)

few carnivores herbivores many plants

Inverted: 1 tree → 1000s of insects

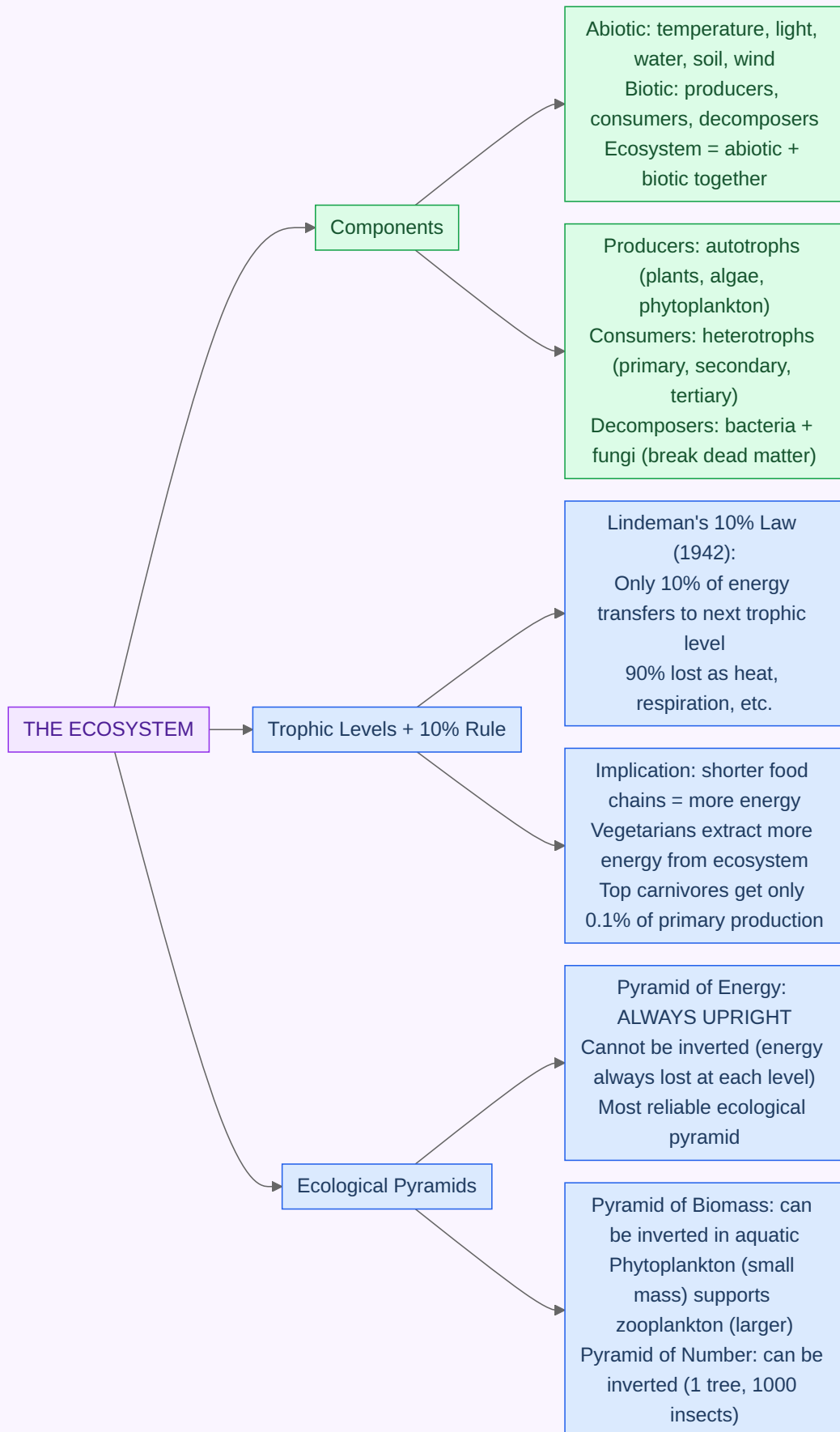
Biomass Pyramid (aquatic = inverted)

zooplankton (large biomass) phytoplankton (small biomass) aquatic inversion: producers reproduce faster than consumed

EXAM ALERT

"Which ecological pyramid is always upright?" — **Pyramid of energy**. This is a standard exam question. The energy pyramid can never be inverted because energy is always lost as heat at each trophic level.

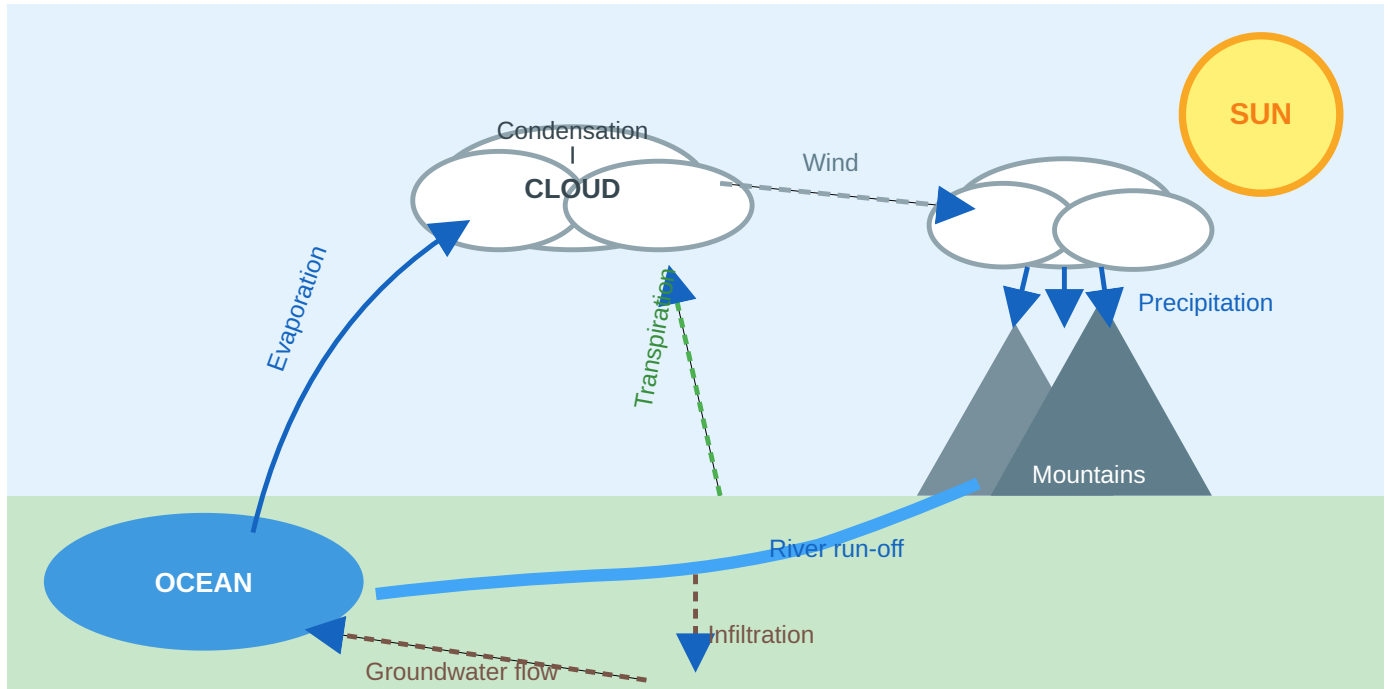
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Chapter A2 — Biogeochemical Cycles

Water Cycle

Evaporation from oceans → condensation into clouds → precipitation (rain/snow) → runoff into rivers → infiltration into groundwater → back to ocean. About 97% of Earth's water is in the oceans (saline). Only ~2.5% is freshwater, and most of that is locked in ice caps and glaciers.



Key facts: ~97% water in oceans | Hydrological cycle driven by solar energy | Transpiration + Evaporation = Evapotranspiration | Infiltration → aquifers (groundwater). Condensation nuclei = dust particles. India: SW monsoon = 75% of annual rainfall.

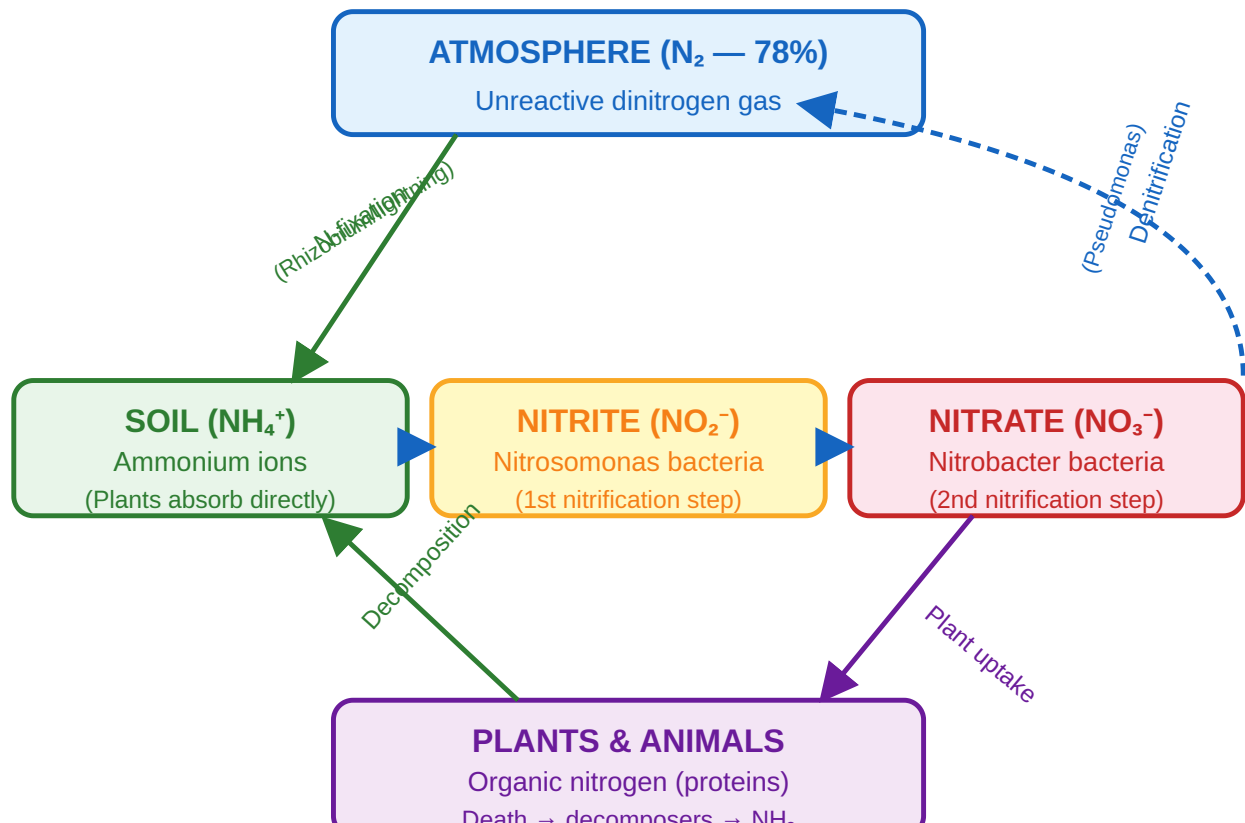
Carbon Cycle

Plants absorb CO_2 through photosynthesis and convert it into biomass. Animals eat plants and release CO_2 through respiration. Decomposers break down dead matter, releasing CO_2 . Fossil fuels are ancient biomass — burning them releases carbon that had been locked underground for millions of years, short-circuiting the natural cycle.

Nitrogen Cycle

Atmosphere is 78% N_2 , but plants cannot use it directly. The cycle works through:

- **Nitrogen fixation** — converts N_2 to NH_3 usable by plants. Done by Rhizobium (in legume root nodules), Azotobacter (free-living in soil), cyanobacteria (Anabaena, Nostoc), lightning (abiotic), and the Haber-Bosch industrial process.
- **Nitrification** — bacteria convert $\text{NH}_3 \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^-$ (usable by plants).
- **Denitrification** — bacteria return NO_3^- to atmospheric N_2 .



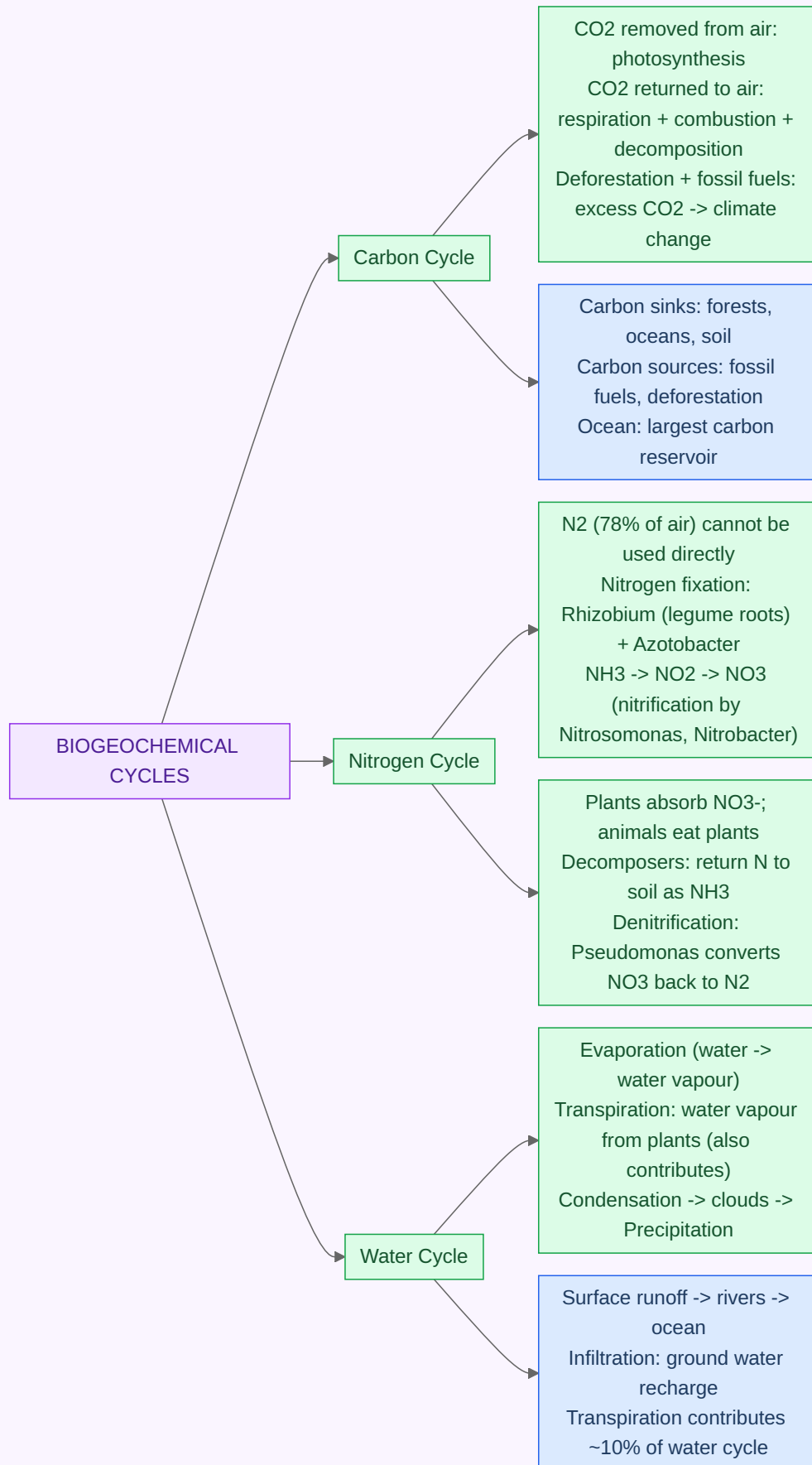
Key bacteria: Rhizobium (fix) | Nitrosomonas (NH₄ → NO₂) | Nitrobacter (NO₂ → NO₃) | Pseudomonas (denitrify)
 Lightning also fixes N₂. Haber–Bosch process = industrial fixation. Legumes host Rhizobium in root nodules.

EXAM ALERT

"Which element's cycle has no gaseous phase?" — **Phosphorus**. Unlike carbon, nitrogen, and sulphur, phosphorus does not enter the gaseous phase. It cycles only through rock → soil → plants → animals → decomposers → back to soil. Phosphorus is often the limiting nutrient in lakes (which is why phosphate-rich fertiliser runoff causes algal blooms).

PART B — BIODIVERSITY

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Chapter B1 — Levels and Importance

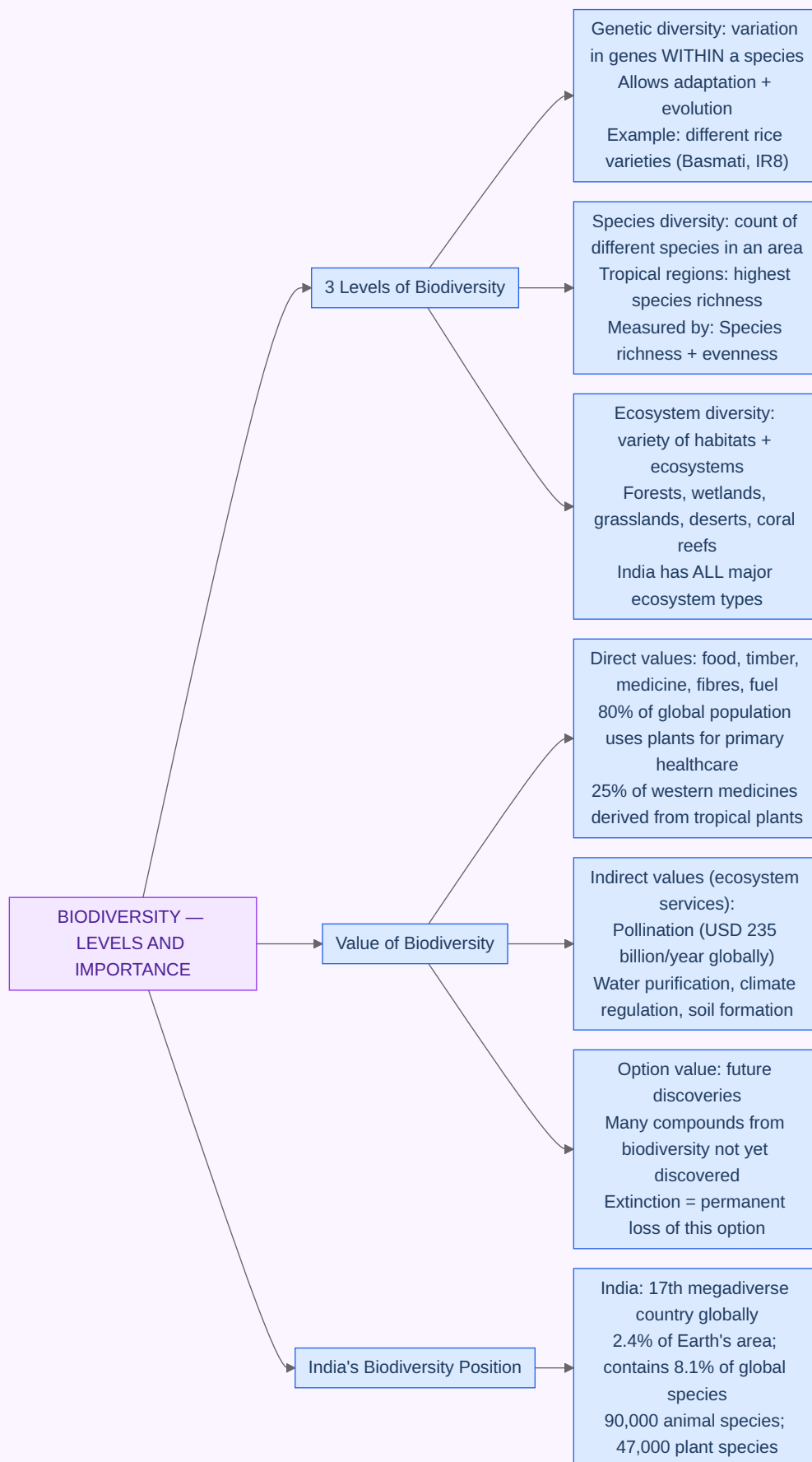
- **Genetic diversity** — variation in genes within a species (e.g., different rice varieties).

- **Species diversity** — variety of species within an ecosystem.
- **Ecosystem diversity** — variety of habitats and ecosystems (forests, wetlands, grasslands).

Why Biodiversity Matters — Ecosystem Services

Category	What we get
Provisioning	Food, water, timber, medicine, raw materials
Regulating	Climate regulation, pollination, flood control, disease control
Cultural	Recreation, spiritual value, tourism, aesthetic
Supporting	Nutrient cycling, soil formation, water purification

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Chapter B2 — Biodiversity Hotspots

A biodiversity hotspot must meet two strict criteria: it must have **at least 1500 species of endemic vascular plants**, AND must have lost **at least 70% of its original primary vegetation**. Norman Myers proposed the concept in 1988. There are now **36 global hotspots**.

India has 4 biodiversity hotspots:

1. **Himalayas** — Eastern Himalayan region including Bhutan, parts of Nepal and northeast India.
2. **Indo-Burma** — Northeast India, Myanmar, parts of Bangladesh, southern China.
3. **Western Ghats + Sri Lanka** — Sahyadri range + Sri Lanka.
4. **Sundaland** — India's portion is the Nicobar Islands only.

REMEMBER

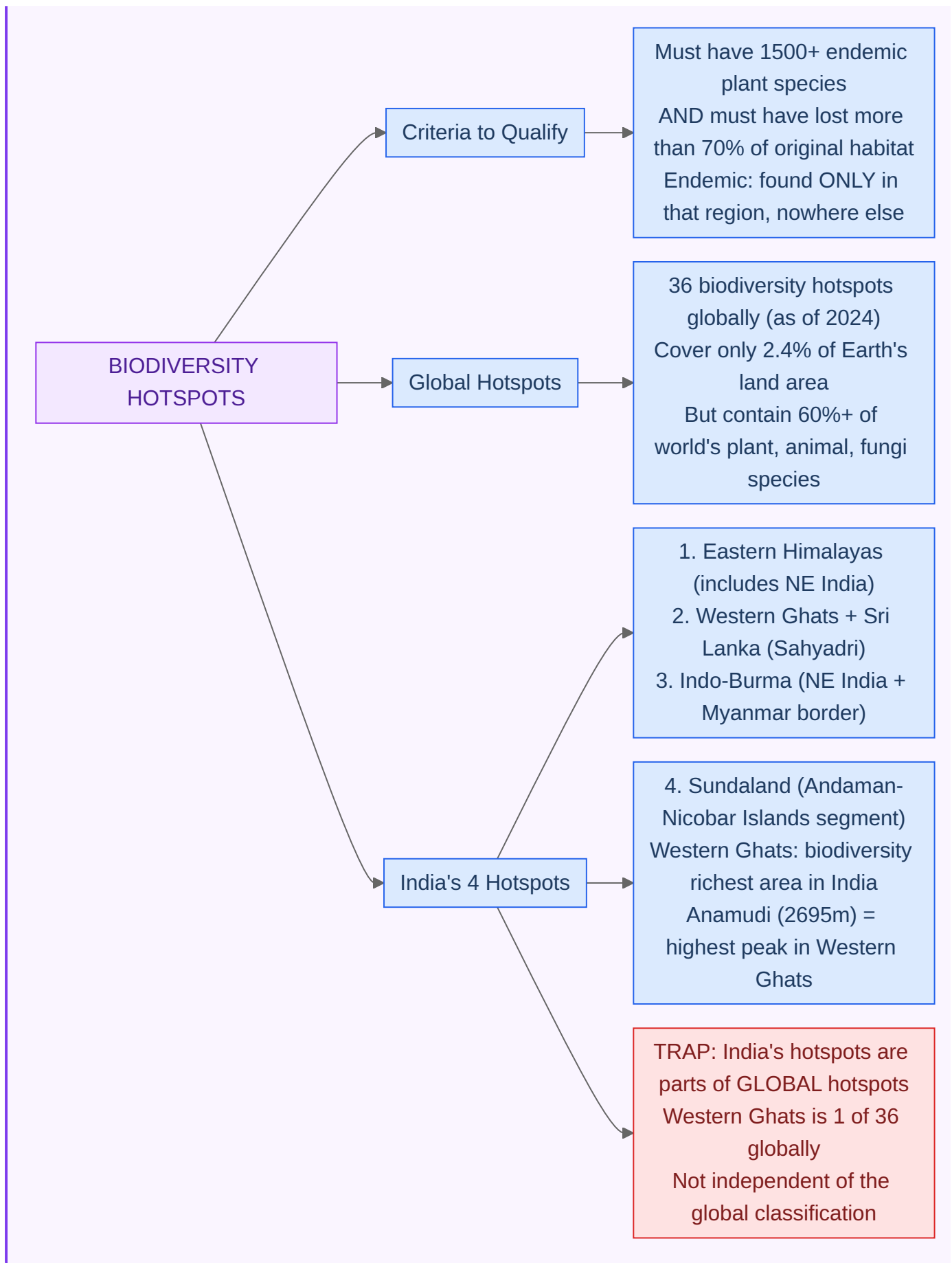
India's 4 hotspots: "**Himalayas, Indo-Burma, Western Ghats, Sundaland**" — **HIWS** — "**H**ot **I**ndian **W**ildlife **S**pot"

India is also one of the **17 megadiverse countries** that together hold approximately 70% of global biodiversity. The others include Brazil, Colombia, Indonesia, Mexico, Australia, China, DRC, and more.

KEY FACT

Key biome memory trick: "**The closer to the equator, the more biodiversity.**" Tropical rainforests (equator) → most species. Temperate forests (mid-latitudes) → fewer. Tundra (polar) → fewest. This single rule correctly answers any "which biome has highest/lowest biodiversity" question. The reason: equatorial regions have year-round warmth, high rainfall, and no seasonal die-off — all three conditions maximise species richness.

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Chapter B3 — IUCN Red List Categories

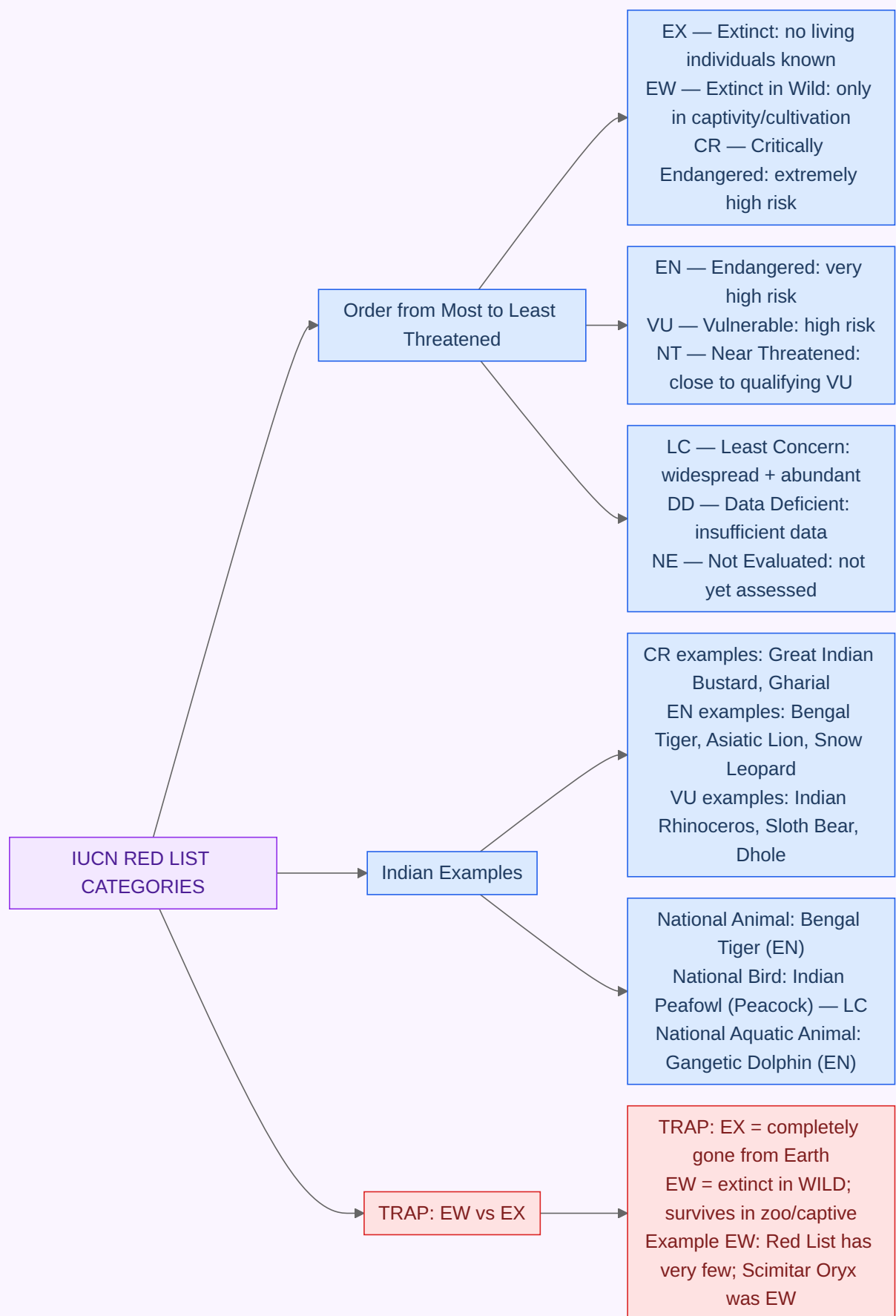
EX (Extinct) → **EW** (Extinct in Wild) → **CR** (Critically Endangered) → **EN** (Endangered) → **VU** (Vulnerable) → **NT** (Near Threatened) → **LC** (Least Concern) → **DD** (Data Deficient) → **NE** (Not Evaluated)

REMEMBER

"Every Experienced Conservationist Encourages Very Nature-Loving Decisions" — EX, EW, CR, EN, VU, NT, LC, DD (ignore NE).

IUCN (International Union for Conservation of Nature) was established in **1948** and maintains the Red List. The Great Indian Bustard is **Critically Endangered**. The Gangetic Dolphin is **Endangered**.

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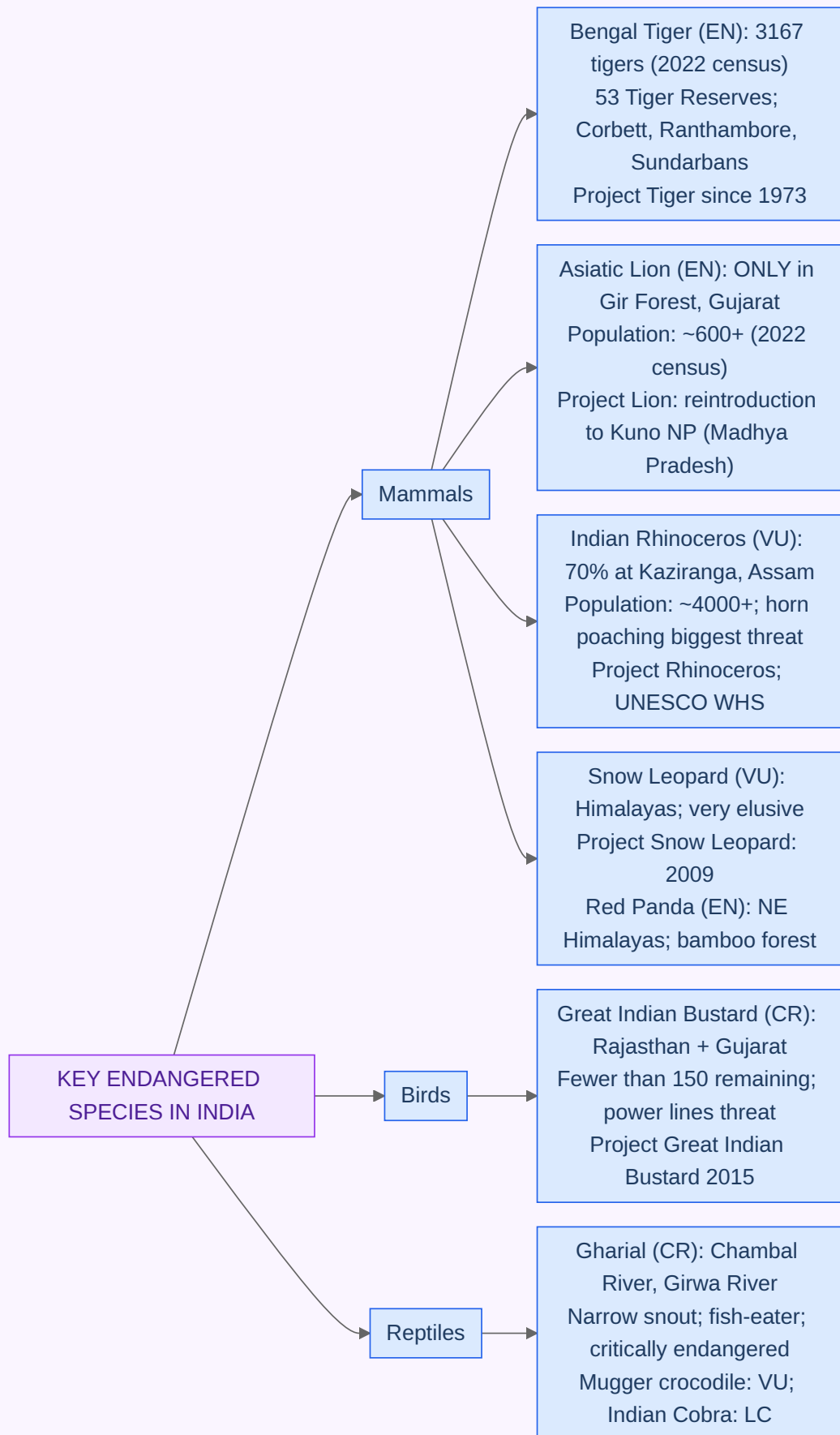
Chapter B4 — Key Endangered Species in India

Species	IUCN Status	Where found
Great Indian Bustard	Critically Endangered	Rajasthan (Desert NP), Gujarat
Hangul (Kashmir Stag)	Critically Endangered	Dachigam NP, J&K
Gharial	Critically Endangered	Chambal river (UP-MP-Rajasthan border)
Indian Vulture (long-billed)	Critically Endangered	Various states (crashed due to diclofenac)
Asiatic Lion	Endangered	Gir NP, Gujarat — the ONLY place in the world
Royal Bengal Tiger	Endangered	Multiple PAs; Sundarbans, Ranthambore, Kanha
Indian Elephant	Endangered	Western Ghats, northeast India
Sangai (brow-antlered deer)	Endangered	Keibul Lamjao, Manipur
Nilgiri Tahr	Endangered	Western Ghats (Eravikulam NP)
One-horned Rhinoceros	Vulnerable	Kaziranga, Manas (Assam)
Snow Leopard	Vulnerable	Hemis (Ladakh), Spiti (HP)
Olive Ridley Turtle	Vulnerable	Gahirmatha (Odisha) — world's largest nesting beach

PART C — PROTECTED AREAS AND INDIA'S WILDLIFE

India has the world's largest tiger population (3,167 — 2022 census) and hosts biodiversity hotspots in the Western Ghats, Eastern Himalayas, Indo-Burma region, and the Sundaland cluster in Andaman. Protected areas are classified under the **Wildlife (Protection) Act 1972** into National Parks, Wildlife Sanctuaries, Conservation Reserves, and Community Reserves. Additionally, UNESCO MAB-recognised Biosphere Reserves and NTCA-governed Tiger Reserves form the conservation backbone. Examiners test the names, states, and legal classification of these areas heavily — especially first/largest/smallest NPs, the UNESCO World Network Biosphere Reserves, and which animals belong to which conservation project.

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Chapter C1 — Types of Protected Areas

Under the **Wildlife (Protection) Act 1972**:

1. **National Park** — highest protection; no human activity (including grazing) allowed inside. India has **106 NPs**.
2. **Wildlife Sanctuary** — lower protection than NP; some regulated activities permitted. India has **~570 WLSs**.
3. **Conservation Reserve** — on community or state land around NPs/WLSs.
4. **Community Reserve** — private or community land managed for conservation.

Tiger Reserves — administered by NTCA under Project Tiger. As of April 2026: **58 Tiger Reserves**.

Biosphere Reserves — UNESCO MAB (Man and Biosphere) programme. India has **18 BRs, of which 12 are on the UNESCO MAB World Network**.

Elephant Reserves — under Project Elephant. India has **33 Elephant Reserves**.

KEY FACT

Mnemonic — NP vs WLS: Think "**No People**" vs "**Wildlife + Limited Settlement**." A National Park = **No** human activity. A Wildlife **S**anctuary = **S**ome regulated use still allowed. If the question says "no grazing, no forestry, no settlement" → National Park. If "regulated grazing allowed" → Wildlife Sanctuary.

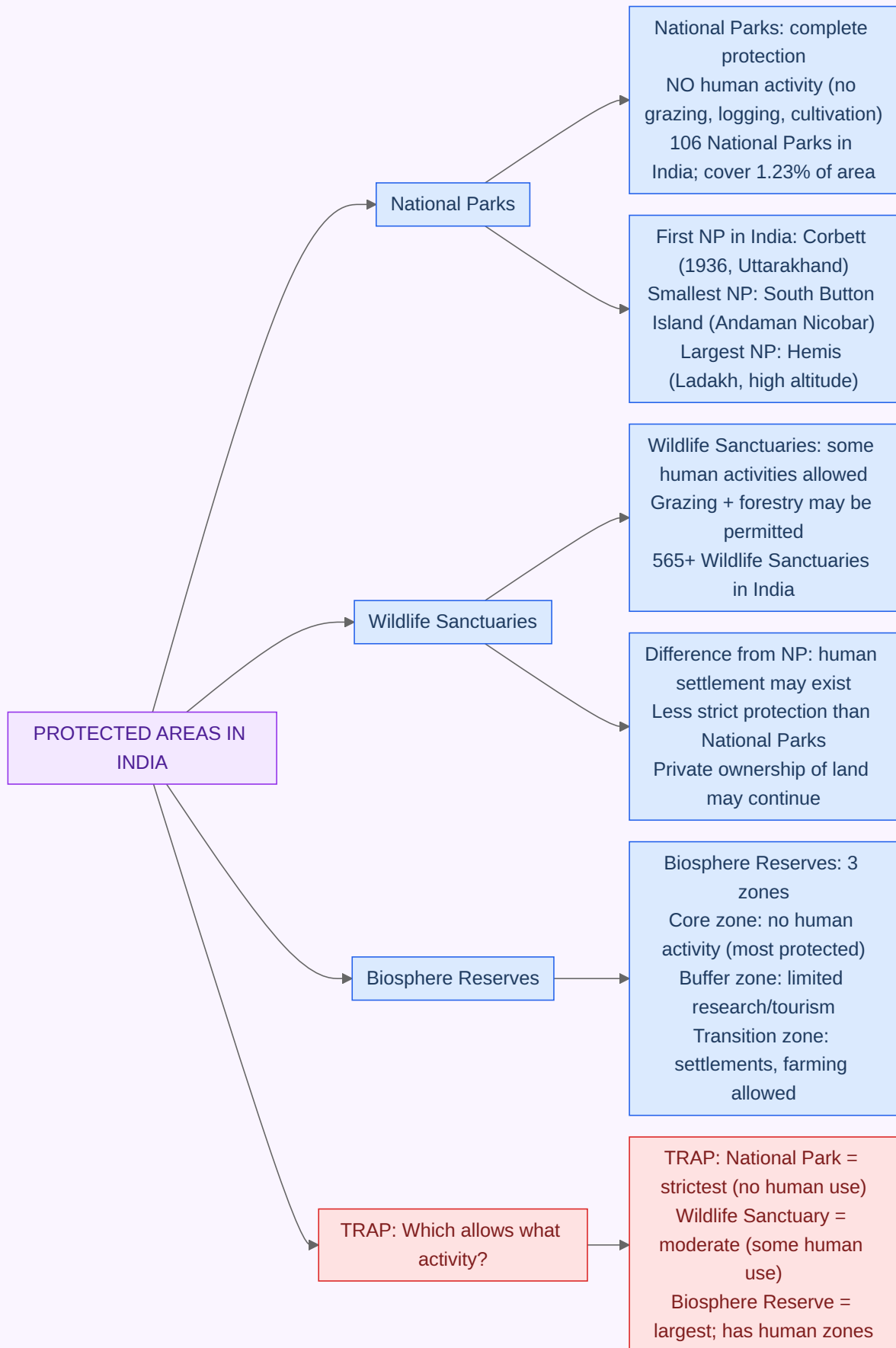
EXAM ALERT

⚠ TRAP — Ramsar vs Biosphere Reserve vs World Heritage Site: candidates mix these every year.

Category	Governing body	Covers	India count
Ramsar Site	Ramsar Convention (1971, Iran)	Wetlands only	100 (June 2026)
Biosphere Reserve	UNESCO MAB Programme	Large landscape with 3 zones	18 (12 on UNESCO list)
UNESCO World Heritage Site	UNESCO WHC	Outstanding universal value	42 (as of 2024)

A site can hold ALL THREE designations simultaneously — Sundarbans is a Ramsar site, a UNESCO WHS, and part of a Biosphere Reserve. Examiners test all three labels on the same site deliberately.

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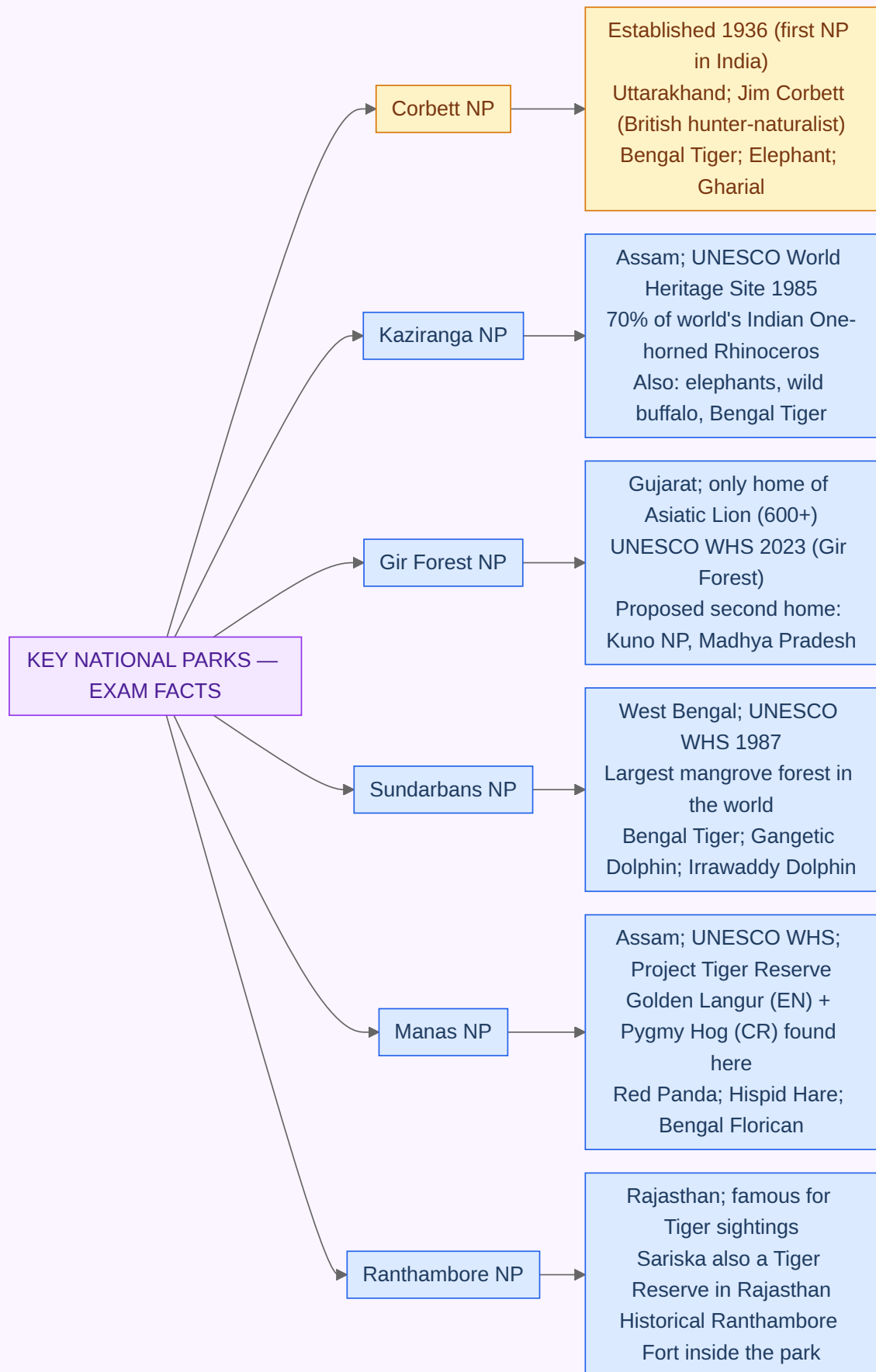
Chapter C2 — Key National Parks — Must-Know Facts

KEY FACT

The most-tested NP facts: **India's first NP = Jim Corbett (originally Hailey NP, 1936, Uttarakhand)**. **Largest NP by area = Hemis (Ladakh, ~4,400 km²)**. **Only floating NP = Keibul Lamjao (Manipur, on Loktak Lake)**. **Only UNESCO Mixed Heritage NP = Khangchendzonga (Sikkim)**. **Only place for Asiatic Lion = Gir (Gujarat)**.

State	Key National Parks
Uttarakhand	Jim Corbett (1st NP, 1936), Nanda Devi + Valley of Flowers (UNESCO), Rajaji, Govind
Rajasthan	Ranthambore (Tiger Reserve), Sariska, Desert NP, Keoladeo Ghana (UNESCO; Siberian cranes)
Gujarat	Gir (Asiatic lion — only habitat in world), Marine NP Gulf of Kachchh (1st marine NP), Velavadar (blackbuck)
Madhya Pradesh	Kanha (barasingha; Mowgli setting), Bandhavgarh (highest tiger density in India), Pench, Panna, Satpura, Madhav
Maharashtra	Tadoba-Andhari (largest in Maharashtra), Sanjay Gandhi (Borivali), Pench
Assam	Kaziranga (UNESCO; one-horned rhinoceros), Manas (UNESCO; pygmy hog)
Karnataka	Bandipur (1st Project Tiger reserve, 1973), Nagarhole, Kudremukh, Bannerghatta
Kerala	Silent Valley, Periyar, Eravikulam (Nilgiri tahr), Anamudi Shola
Jammu & Kashmir / Ladakh	Dachigam (Hangul deer), Hemis (largest NP; snow leopard), Salim Ali
West Bengal	Sundarbans (UNESCO; mangrove tigers), Buxa, Gorumara, Jaldapara, Singalila
Arunachal Pradesh	Namdapha, Mouling
Manipur	Keibul Lamjao (only floating NP; Sangai deer; on Loktak Lake)
Sikkim	Khangchendzonga (only UNESCO Mixed Heritage NP in India)
Himachal Pradesh	Great Himalayan NP (UNESCO), Pin Valley
Odisha	Simlipal (Biosphere Reserve + Tiger Reserve), Bhitarkanika (saltwater crocodile)

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Chapter C3 — Biosphere Reserves (India's 18)

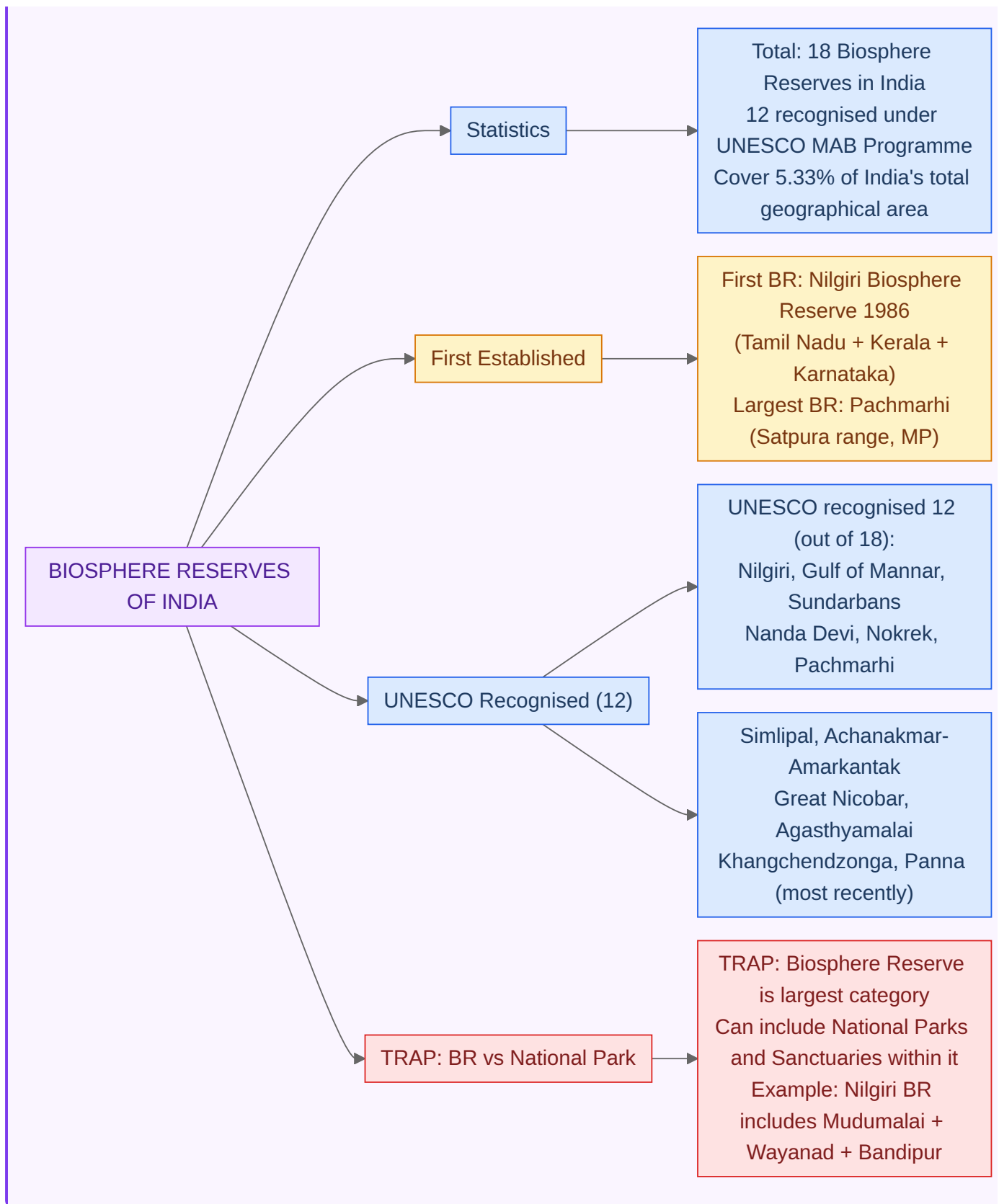
India's 18 BRs (12 are on UNESCO MAB World Network list, marked with *):

Nilgiri (1st, 1986), Nanda Devi, Nokrek, Gulf of Mannar, Sundarbans, Manas, Similipal, Pachmarhi, Khangchendzonga, Agasthyamala, Great Nicobar, Achanakmar-Amarkantak, Dihang-Dibang, Dibru-Saikhowa, Panna (listed 2020), Seshachalam, Cold Desert (HP), Kachchh (Gujarat — largest by area).

EXAM ALERT

India's first Biosphere Reserve = **Nilgiri (1986)**. Largest BR = **Kachchh, Gujarat**. Latest BR to be added to UNESCO MAB list = **Panna (2020)**. Total: 18 BRs, 12 on UNESCO list.

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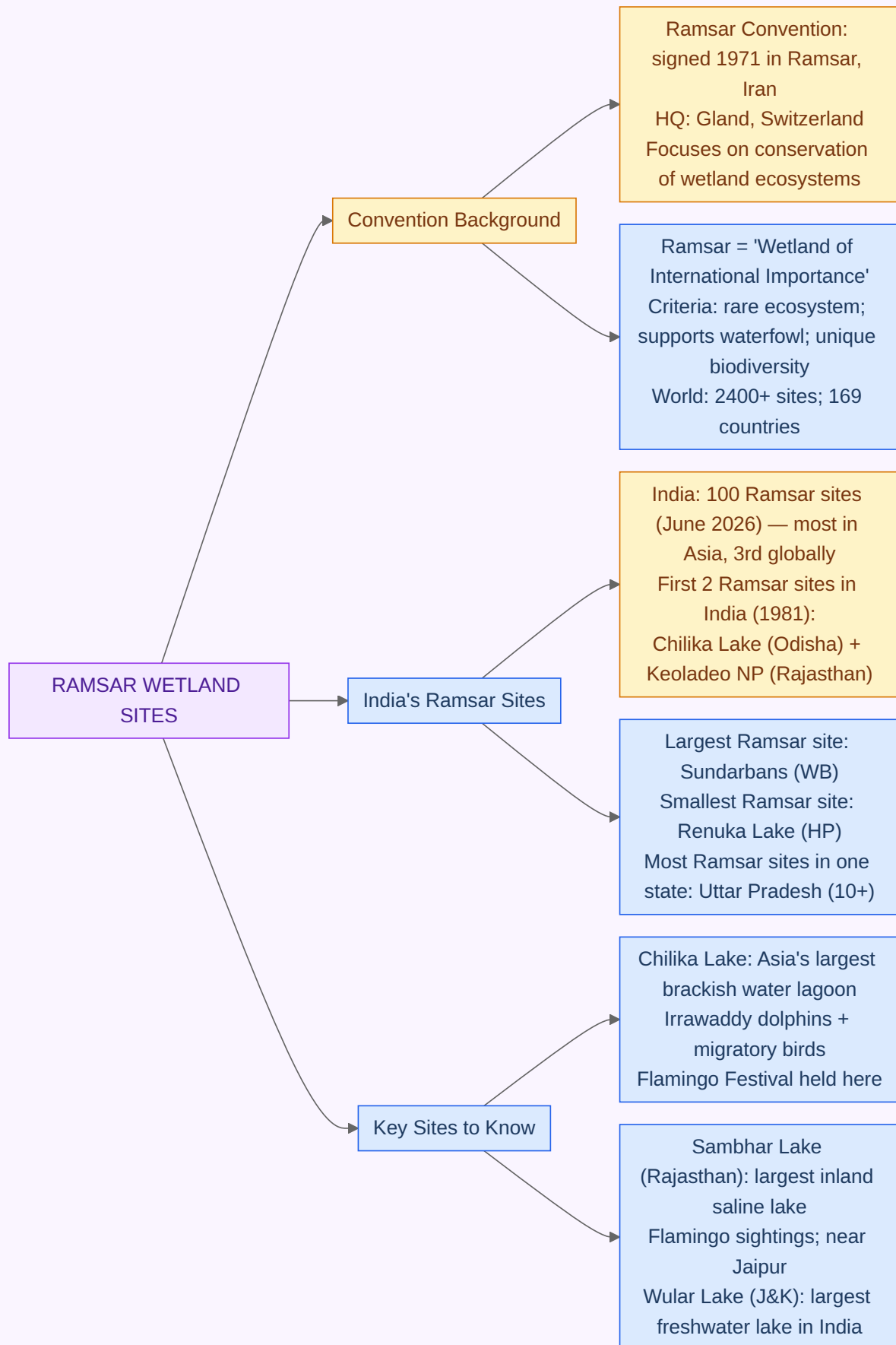
Chapter C4 — Ramsar Sites (Wetlands)

- **Ramsar Convention** — signed at Ramsar, Iran in **1971**; entered force 1975. India ratified in **1982**.
- India's first Ramsar sites: **Chilika (Odisha)** and **Keoladeo Ghana (Rajasthan)**, both designated in **1981** (under the 1982 ratification).
- **100 Ramsar sites** in India as of June 2026 (The 100th was Jai Prakash Narayan Bird Sanctuary, UP) — highest number in Asia and 3rd globally behind UK and Mexico.

- **Tamil Nadu** has the most Ramsar sites at the state level.
- **Largest Ramsar site in India:** Sundarbans Wetland, West Bengal.

PART D — POLLUTION

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Chapter D1 — Air Pollution

Primary and Secondary Pollutants

Primary pollutants (emitted directly): CO, NO_x, SO_x, PM (particulate matter), VOCs, NH₃, heavy metals (Pb, Hg, Cd).

Secondary pollutants (formed by chemical reactions in the atmosphere): - Ground-level ozone (O₃) — formed when NO_x reacts with VOCs in sunlight. - PAN (peroxyacetyl nitrate) — component of photochemical smog. - Acid rain compounds (H₂SO₄, HNO₃) — formed when SO₂ and NO_x dissolve in rain.

KEY FACT

Photochemical smog = NO_x + VOCs + sunlight → ground-level O₃ + PAN. First observed in Los Angeles. London smog (1952) was a different type — classical smog from SO₂ + coal burning + fog = "pea soup" fog. They are different problems with different causes.

Particulate Matter — PM2.5 and PM10 (must-know definitions)

KEY FACT

PM2.5 — particles with diameter **less than 2.5 µm (micrometres)**. So fine they penetrate deep into the lungs and enter the bloodstream. The most dangerous particulate for health; linked to heart disease, stroke, lung cancer.

PM10 — particles with diameter **less than 10 µm**. Enter the lungs but generally stopped before reaching the bloodstream.

Why PM2.5 is worse: It is ~30× smaller than a human hair. No filter in the nose or throat can stop it.

National Ambient Air Quality Standards (NAAQS) — India (CPCB): - PM2.5 annual standard: **40 µg/m³** (India) vs **5 µg/m³** (WHO 2021 guideline — 8× stricter) - PM10 annual standard: **60 µg/m³** (India)

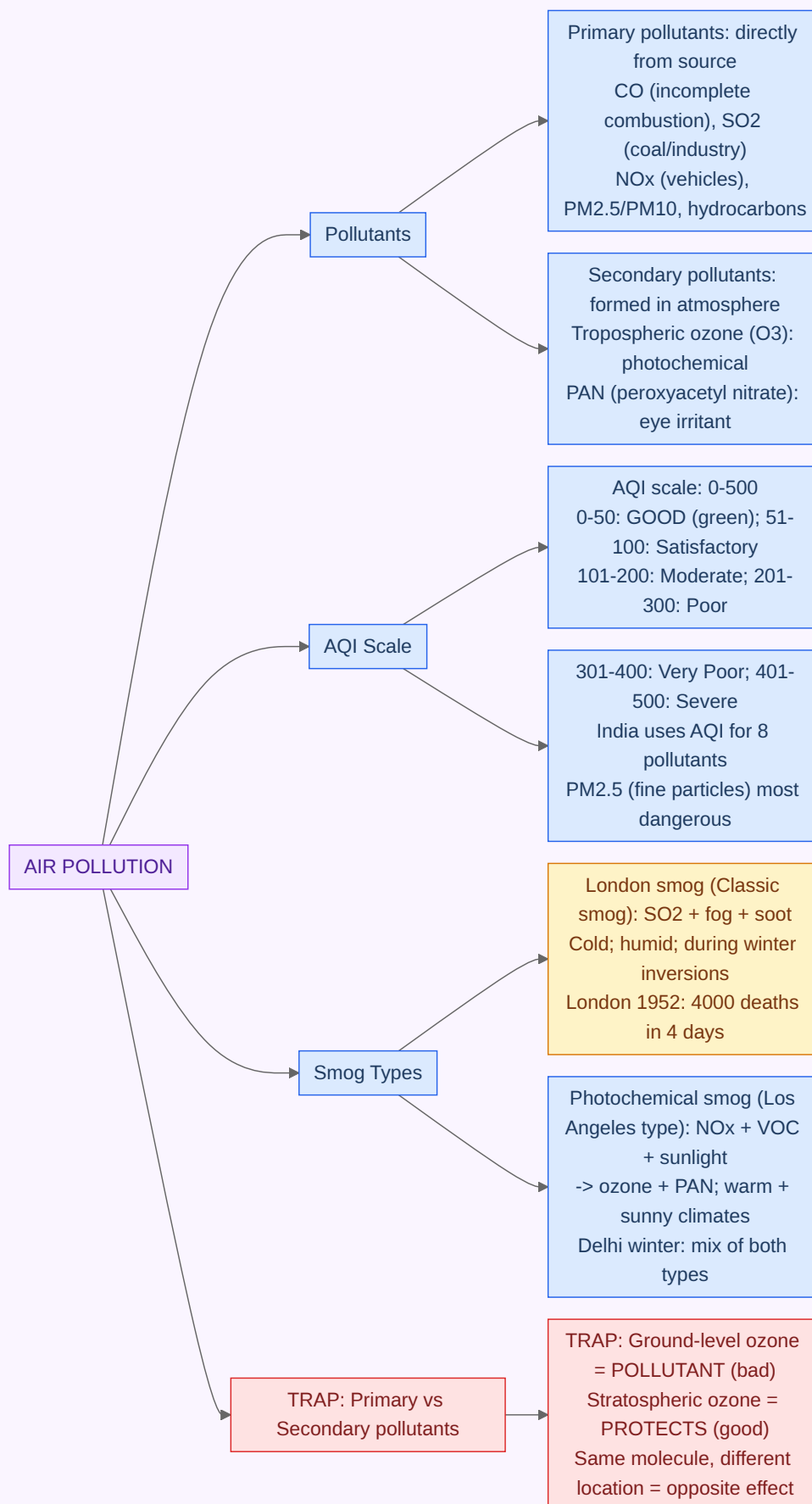
Most Indian cities exceed even the relaxed Indian standard, which is why India appears at the top of global air pollution rankings.

Air Quality Index (India — CPCB)

AQI	Category	Health implication
0–50	Good	Minimal impact
51–100	Satisfactory	Minor breathing discomfort for sensitive people
101–200	Moderate	Breathing discomfort — asthma, heart patients
201–300	Poor	Breathing discomfort for most; avoid outdoor exercise
301–400	Very Poor	Respiratory illness on prolonged exposure
401–500	Severe / Hazardous	Affects healthy people; serious for the sick

AQI is maintained by CPCB (Central Pollution Control Board) and monitored for PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃, NH₃, and Pb.

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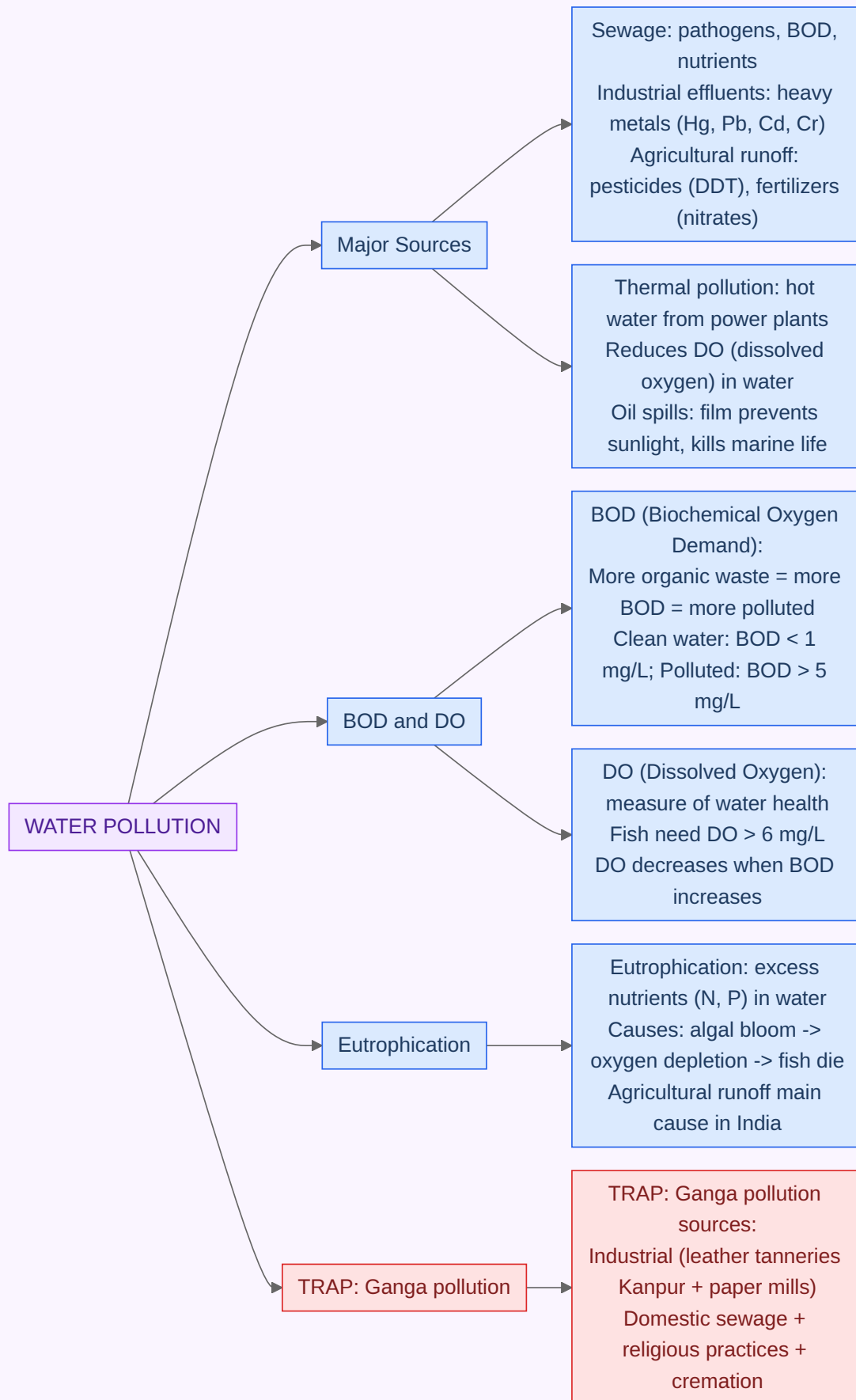
Chapter D2 — Water Pollution

- **BOD (Biological Oxygen Demand)** — the amount of oxygen bacteria need to decompose organic matter in water. High BOD = more organic pollution, less oxygen for fish.
- **DO (Dissolved Oxygen)** — fish need at least 4 mg/L of dissolved oxygen to survive. Polluted water can have near-zero DO.
- **Eutrophication** — excess nutrients (nitrogen, phosphorus from fertiliser runoff) → algal bloom → algae die → bacteria decompose them, consuming all oxygen → fish die. This cycle kills entire lake ecosystems.
- **Bioaccumulation** — toxic substances (DDT, Hg) build up in one organism.
- **Biomagnification** — concentration of toxic substances increases at each trophic level. Top predators (eagles, large fish) suffer the most.

Diseases from Water Pollution

Disease / Condition	Cause
Minamata disease	Mercury (Hg) poisoning — Minamata Bay, Japan (1956)
Itai-itai disease	Cadmium (Cd) poisoning — Jinzu River, Japan
Blue baby syndrome	High nitrates in water → methemoglobinemia in infants
Fluorosis	Fluoride >1.5 ppm → mottled teeth, skeletal damage
Arsenicosis	Arsenic in groundwater — severe in West Bengal and Bangladesh

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Chapter D2B — Ozone Layer

What is the ozone layer?

- **Ozone (O_3)** is a molecule of three oxygen atoms.
- In the **stratosphere** (15–50 km altitude), ozone forms a protective layer that absorbs 97–99% of the Sun's harmful **ultraviolet (UV-B and UV-C) radiation**.
- Without this layer, UV radiation would cause massive increases in skin cancer, cataracts, and damage to crops and marine ecosystems.

EXAM ALERT

Exam trap — where is the ozone layer?

"The ozone layer is in which part of the atmosphere?" → **Stratosphere** (NOT troposphere).

Ground-level ozone = bad (pollutant). Stratospheric ozone = good (protects life). Same molecule; opposite roles; different altitude. Examiners test this distinction every year.

Ozone Hole — Discovery and Cause

- The **ozone hole** forms primarily over **Antarctica** (and to a lesser extent the Arctic) during spring.
- **Discovered in 1985** by British scientists **Farman, Gardiner, and Shanklin** (British Antarctic Survey).
- **Cause: CFCs (chlorofluorocarbons)** — used in refrigerators, air conditioners, aerosol sprays — rise into the stratosphere, break down under UV radiation, and release chlorine atoms. One chlorine atom can destroy **100,000 ozone molecules**.

Montreal Protocol (1987) — Most Successful Environmental Treaty

- **Signed:** 1987, entered force 1989.
- **Goal:** Phase out ozone-depleting substances (ODS) — primarily CFCs, HCFCs, halons.
- Regarded as the **most successful global environmental treaty** because:
- Near-universal ratification (197 countries).
- Ozone hole is slowly recovering — projected full recovery by ~2060–2070.
- India joined in **1992**.

Kigali Amendment (2016) to Montreal Protocol

- **Agreed:** October 2016, Kigali, Rwanda. Entered force 2019.
- **What it does:** Adds phase-down of **HFCs (hydrofluorocarbons)** to the Montreal Protocol.
- **Why important:** HFCs were introduced as replacements for CFCs (they do NOT deplete ozone) — but they are **potent greenhouse gases** with GWP of hundreds to thousands.
- By phasing down HFCs (mainly used in ACs and refrigerators), Kigali Amendment addresses **both** ozone protection AND climate change.
- India ratified in **2021**.

EXAM ALERT

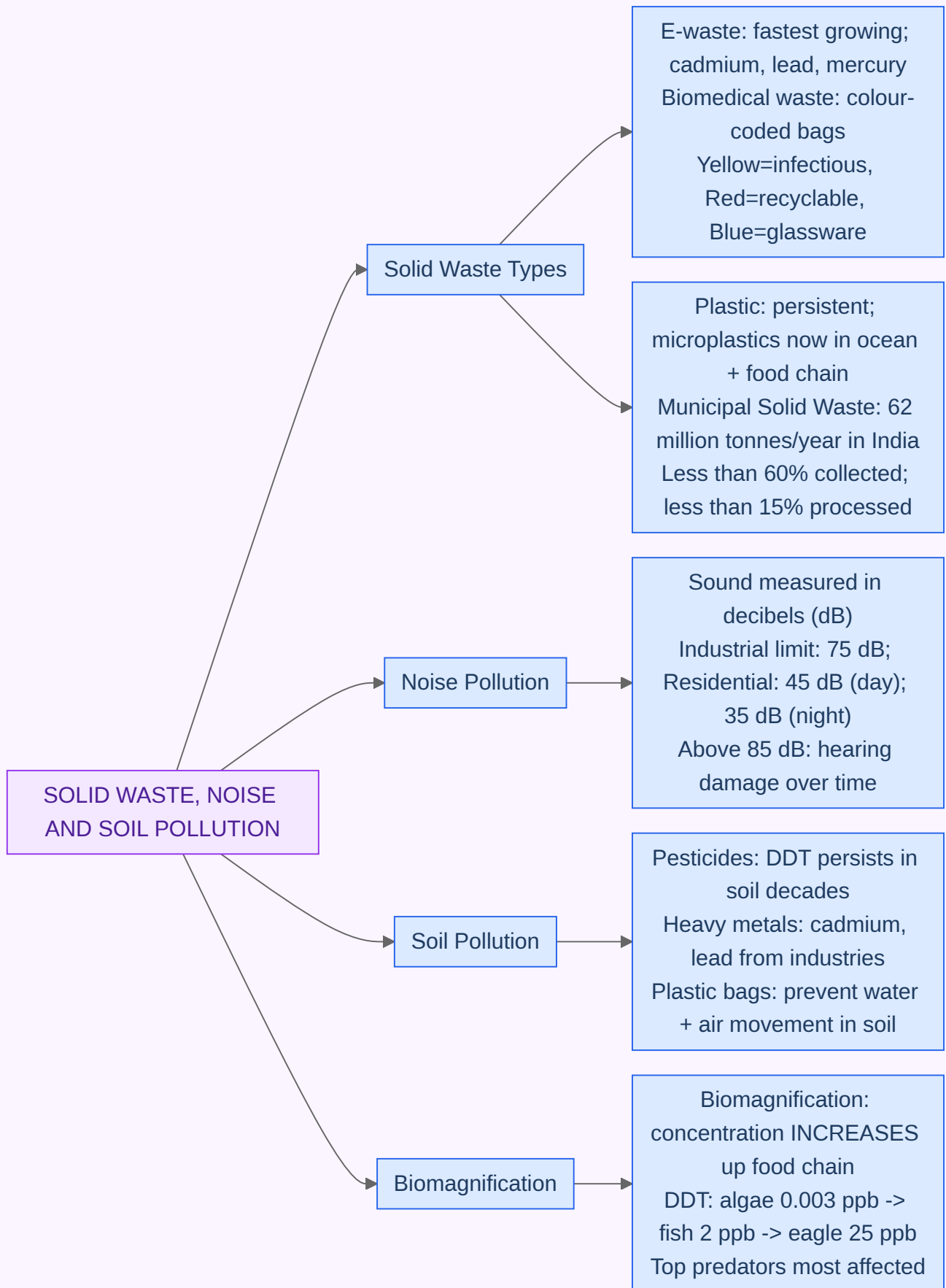
Kigali Amendment exam trap: "What does the Kigali Amendment target?" → **HFCs** (not CFCs — those are already phased out). HFCs were the CFC replacements but are powerful GHGs. Kigali closes this loophole.

Chapter D3 — Solid Waste, Noise, Soil

- **E-waste:** India generates ~3.2 million tonnes/year. E-Waste Management Rules 2022 introduced EPR.
- **Plastic:** India banned 19 categories of single-use plastics from **1 July 2022**.
- **Noise pollution:** Measured in decibels (dB). Residential area limit: 55 dB (day), 45 dB (night). Hospital/school silence zones: 50 dB (day), 40 dB (night).
- **Soil pollution:** Pesticides (DDT, organophosphates), industrial heavy metals, plastic litter, e-waste.
- **Persistent Organic Pollutants (POPs)** — DDT, dioxins. Covered by the **Stockholm Convention (2001)**.

PART E — CLIMATE CHANGE

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Chapter E1 — Greenhouse Gases

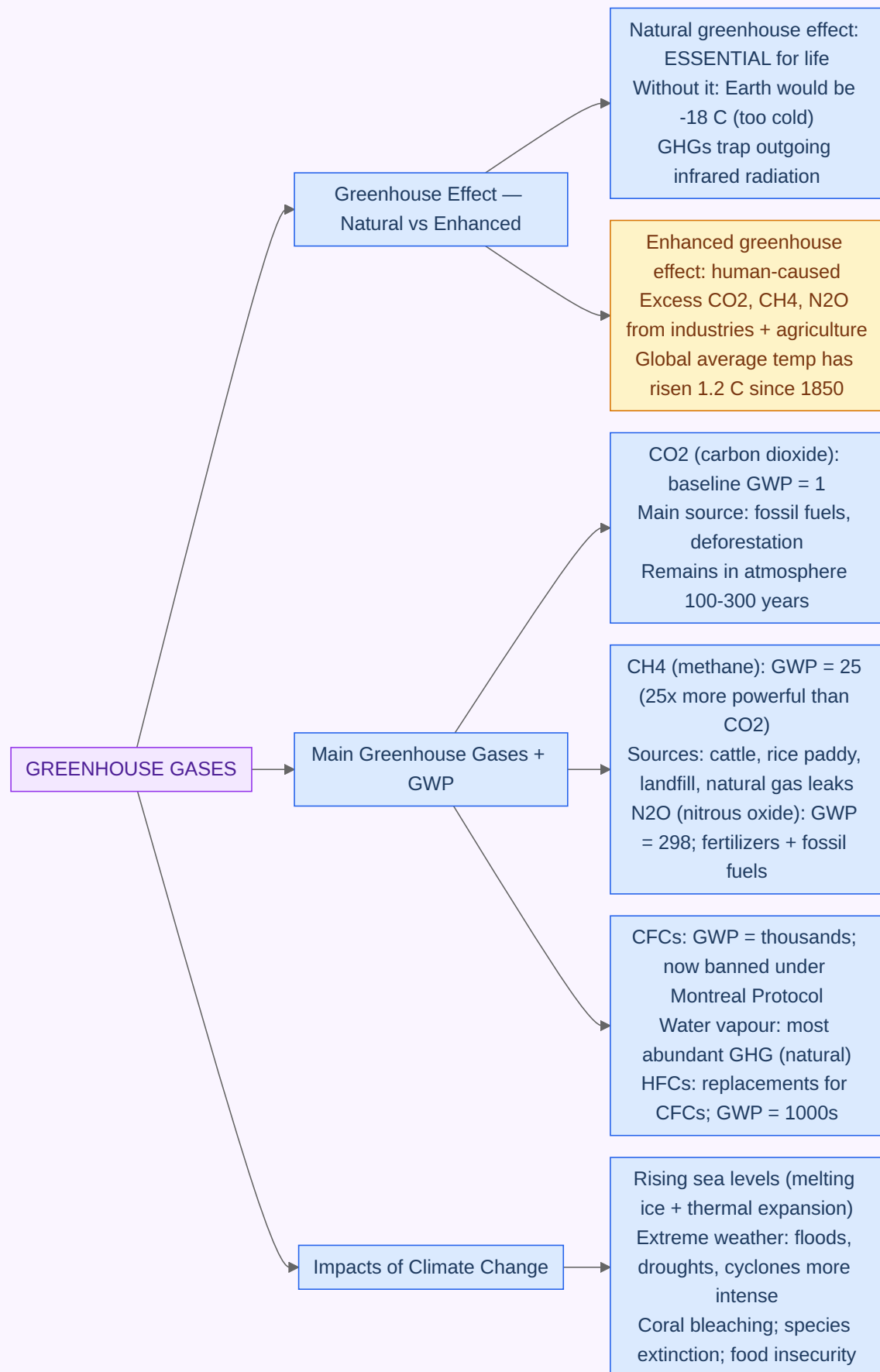
The natural greenhouse effect keeps Earth ~15 °C warmer than it would otherwise be — without it, Earth's average temperature would be -18 °C. The problem is human-enhanced greenhouse warming.

Greenhouse Gas	GWP (100-year, relative to CO ₂)	Main sources
CO ₂	1 (reference)	Fossil fuel combustion, deforestation, cement
CH ₄ (methane)	28–36	Livestock (enteric fermentation), paddy farming, landfills
N ₂ O	298	Fertilisers, industrial processes
HFCs	Hundreds to thousands	Refrigerants, AC units
PFCs	Thousands	Aluminium smelting
SF ₆	23,500	Electrical switchgear — the strongest GHG

EXAM ALERT

The main greenhouse gas by quantity = CO₂. The most potent greenhouse gas by GWP = SF₆ (sulphur hexafluoride, 23,500× CO₂). Examiners test both. Also: "Most abundant greenhouse gas in the atmosphere" = water vapour — but in exam context, CO₂ is the standard answer for the climate-forcing one.

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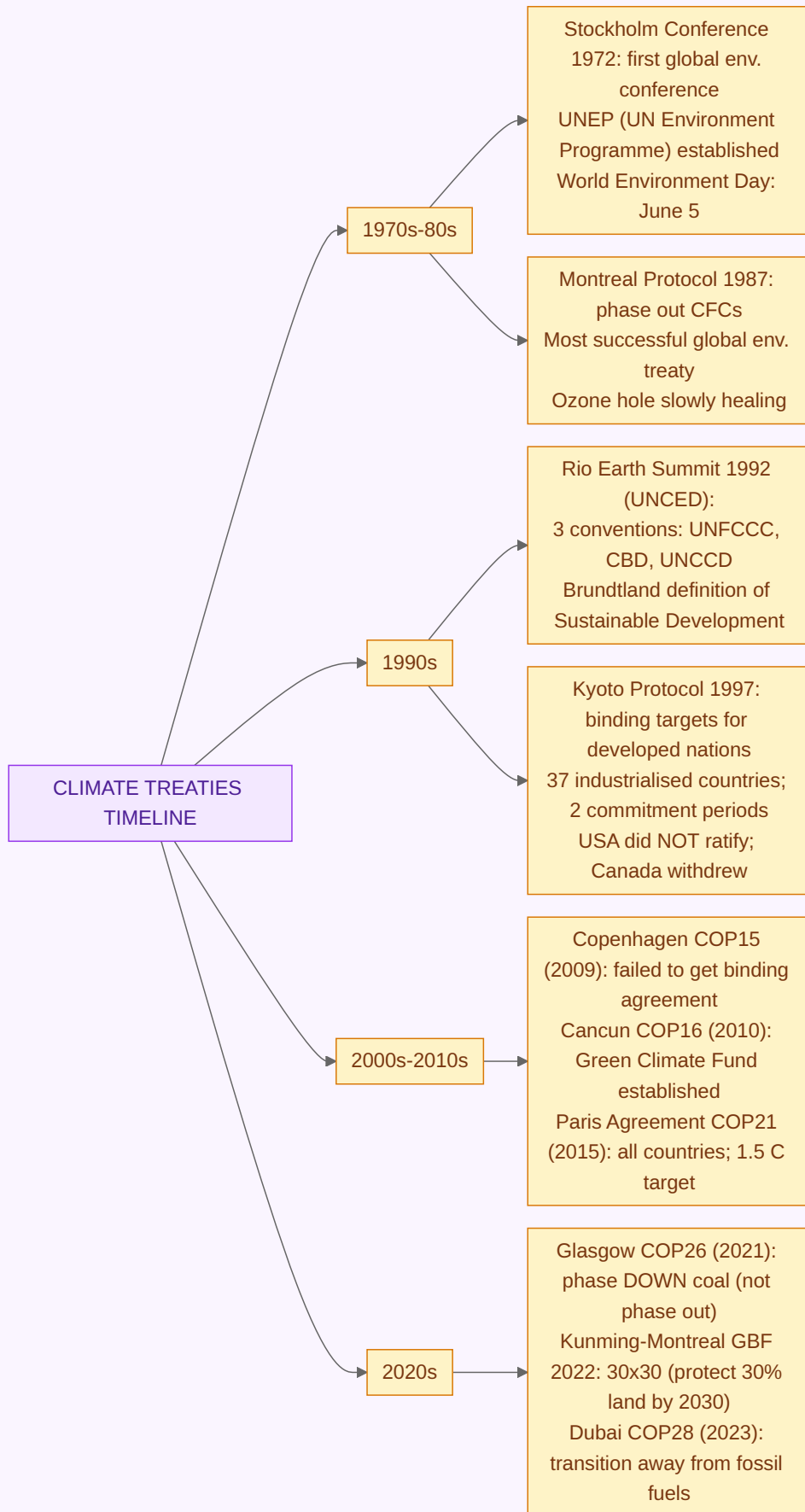
Chapter E2 — Climate Treaties Timeline

Year	Event	Key Outcome
1972	Stockholm Conference on Human Environment	Birth of UNEP (HQ Nairobi); first major UN env conference
1985	Vienna Convention	Framework for ozone layer protection
1987	Montreal Protocol	Phase-out of CFCs and other ODSs; widely regarded as most successful env treaty
1988	IPCC formed	WMO + UNEP; Nobel Peace Prize 2007
1992	Rio Earth Summit (UNCED)	UNFCCC, CBD, Agenda 21 signed
1997	Kyoto Protocol	Binding GHG cuts on developed (Annex-I) countries; India = non-Annex-I (no binding cuts)
2009	Copenhagen COP-15	Failed to achieve binding agreement; voluntary pledges only
2015	Paris Agreement (COP-21)	Limit warming to "well below 2°C, pursue 1.5°C"; voluntary NDCs by all countries
2016	Kigali Amendment (to Montreal)	Phase-down of HFCs
2021	Glasgow COP-26	India's PANCHAMRIT; coal phase-down language
2022	Sharm el-Sheikh COP-27	Loss and Damage fund agreed
2022	Kunming-Montreal GBF (CBD COP-15)	"30×30" — protect 30% of land + sea by 2030
2023	Dubai COP-28	First Global Stocktake; "transition away from fossil fuels"
2024	Baku COP-29	\$300 billion/year climate finance pledge by 2035
2025	Belém COP-30	Brazil host; first COP in the Amazon

COMMON TRAP

Common traps: Vienna Convention (1985) vs Montreal Protocol (1987) — Vienna is the framework, Montreal is the actual action. Kyoto Protocol (1997) — binding only on developed countries. Paris Agreement (2015) — voluntary NDCs for all. Stockholm Convention (2001, about POPs) is separate from Stockholm Conference (1972, about human environment). These are four different things that examiners deliberately mix up.

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Chapter E3 — India's Climate Action

India's PANCHAMRIT (Glasgow COP-26, 2021) — 5 Commitments

1. Achieve **500 GW non-fossil installed capacity** by 2030.
2. **50% of energy needs** from renewables by 2030.
3. Reduce projected carbon emissions by **1 billion tonnes** by 2030.
4. Reduce **emission intensity of GDP by 45%** from 2005 level by 2030.
5. **Net-zero by 2070.**

KEY FACT

India's net-zero target is **2070** — not 2050 or 2060 like most developed countries. This is important for exams. The logic: India is still developing and needs fossil fuels during its growth phase; 2070 gives 20-40 extra years compared to developed nations.

NAPCC — National Action Plan on Climate Change (2008) — 8 Missions

1. National Solar Mission
2. National Mission for Enhanced Energy Efficiency
3. National Mission on Sustainable Habitat
4. National Water Mission
5. National Mission for Sustaining the Himalayan Ecosystem
6. National Mission for a Green India
7. National Mission for Sustainable Agriculture
8. National Mission on Strategic Knowledge for Climate Change

EXAM ALERT

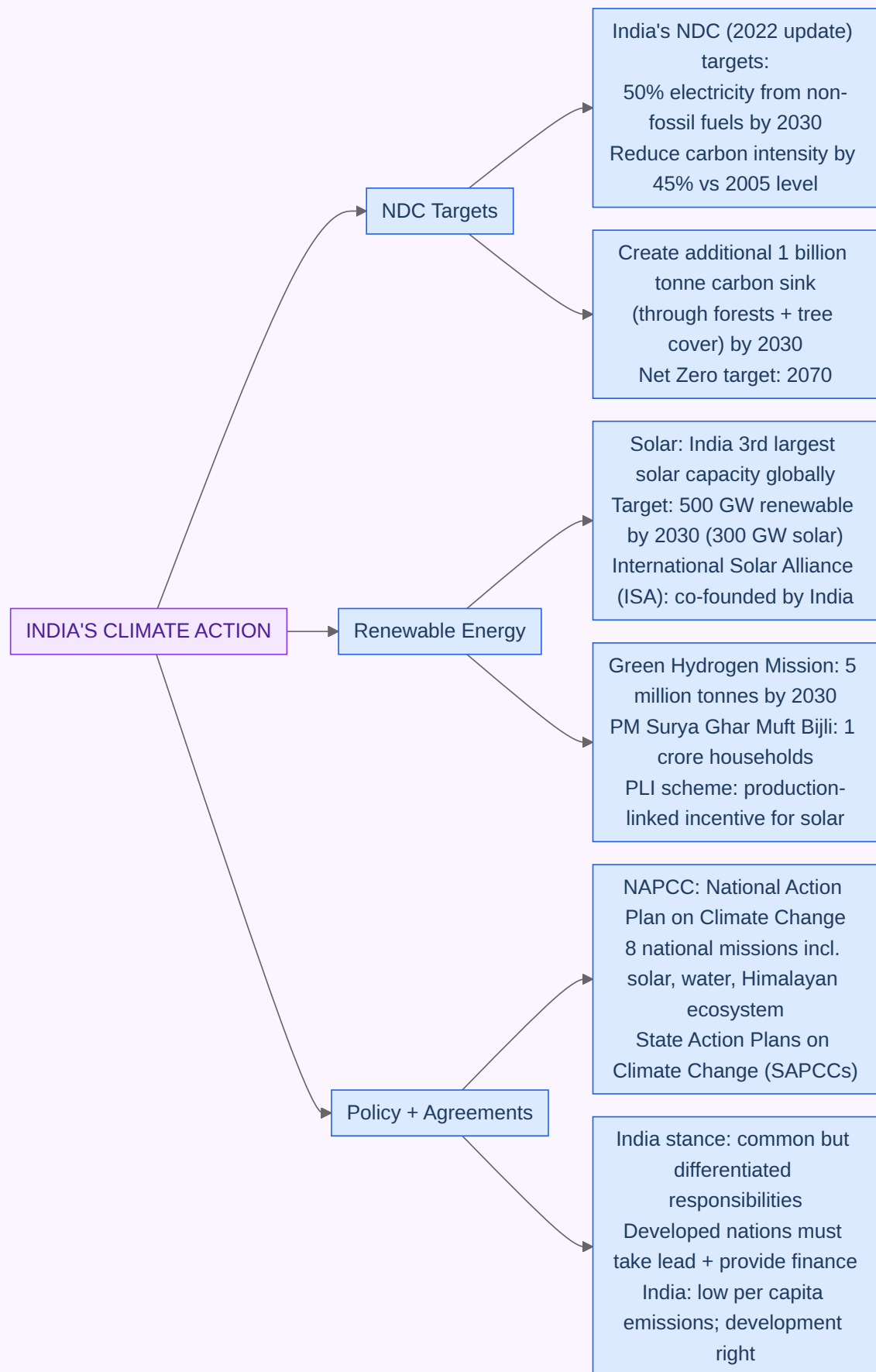
NAPCC has exactly **8 missions**. Examiners often ask the count. The Solar Mission (1st) and Green India Mission (6th) are the most frequently referenced.

Other Key Climate Initiatives

- **International Solar Alliance (ISA)** — launched 2015 by India and France at Paris COP-21; HQ at Gurugram, India.
- **Coalition for Disaster Resilient Infrastructure (CDRI)** — launched by India at UN Climate Action Summit, New York, 2019.
- **Mission LiFE (Lifestyle for Environment)** — launched 2022 at COP-26 by PM Modi; focuses on individual behaviour change.
- **Green Hydrogen Mission (2023)** — target: 5 million metric tonnes of green hydrogen by 2030.
- **FAME-II** — Faster Adoption and Manufacturing of Electric Vehicles.

PART F — INDIAN ENVIRONMENTAL LAWS AND INSTITUTIONS

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



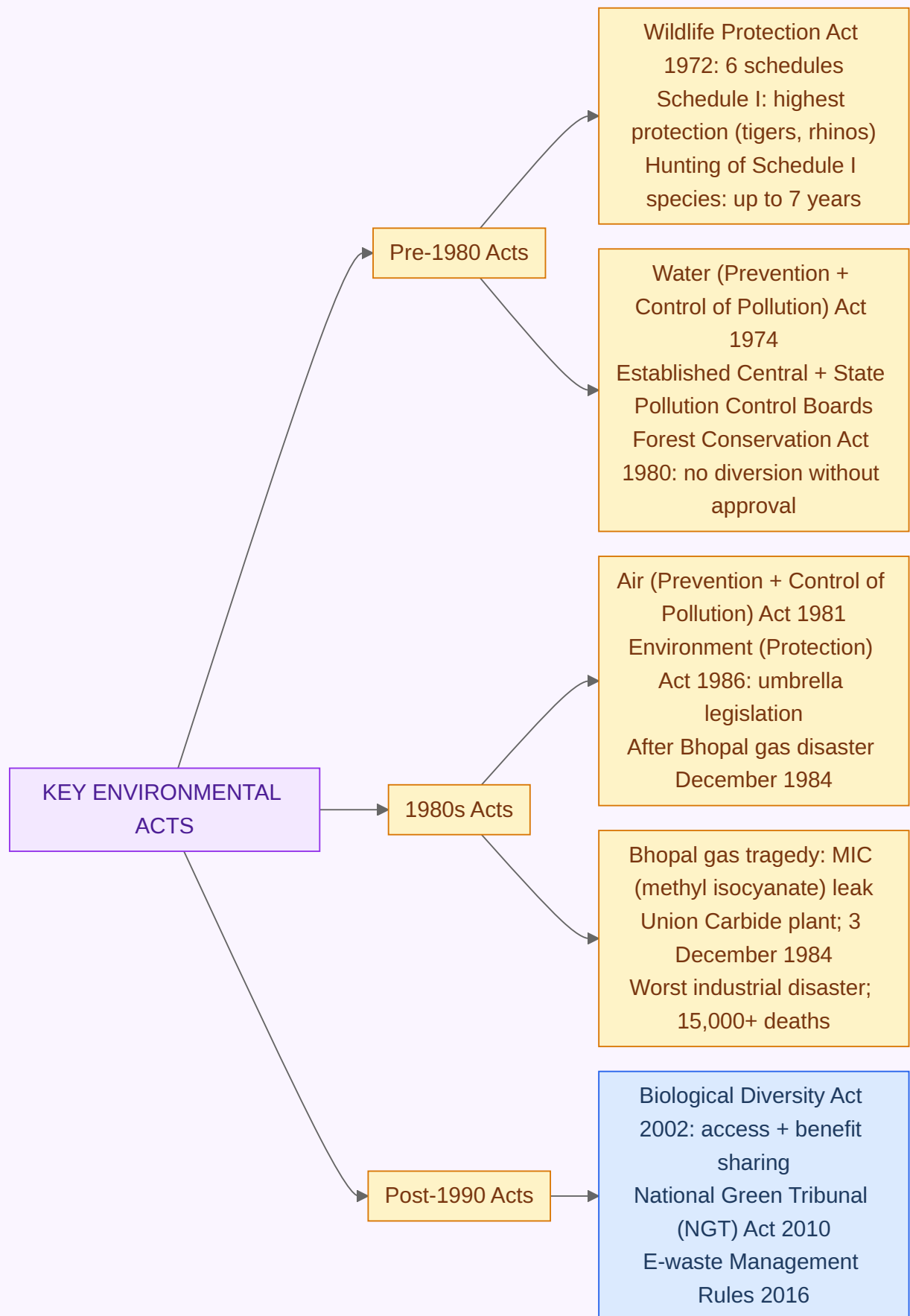
Chapter F1 — Key Environmental Acts (Chronological)

Act	Year	What it does
Indian Forest Act	1927	British-era; defines reserved, protected, village forests
Wildlife (Protection) Act	1972	Framework for protected areas; originally 6 schedules, now 4 after 2022 amendment
Water (Prevention & Control of Pollution) Act	1974	Created CPCB and SPCBs
Forest (Conservation) Act	1980	Diversion of forest land for non-forest purposes requires Central Government approval
Air (Prevention & Control of Pollution) Act	1981	Extended CPCB's mandate to air quality
Environment (Protection) Act	1986	"Umbrella Act" — passed after Bhopal gas tragedy; empowers Centre to issue all environment-related rules
Public Liability Insurance Act	1991	No-fault compensation for industrial accidents involving hazardous substances
Biological Diversity Act	2002	Three-tier system: NBA + SBBs + BMCs
Forest Rights Act	2006	Rights of Scheduled Tribes and Other Traditional Forest Dwellers
National Green Tribunal (NGT) Act	2010	Specialised quasi-judicial body for fast disposal of environmental cases; principal bench in Delhi
Compensatory Afforestation Fund (CAMPA) Act	2016	Manages funds for afforestation when forests are diverted
Wildlife Protection Amendment Act	2022	Aligned with CITES; reduced schedules from 6 to 4
Forest Conservation Amendment Act	2023	Eased clearance for strategic/border projects

COMMON TRAP

The **Environment Protection Act is from 1986**, NOT 1984 (the year of the Bhopal tragedy). The tragedy caused the government to pass the Act in 1986 — two years later. This distinction trips up many students.

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F2 — Wildlife Protection Act 1972 — Schedules

After the 2022 Amendment, WLPA has **4 schedules** (reduced from the original 6):

- **Schedule I** — highest protection; hunting penalty up to 7 years + heavy fine. Includes tiger, lion, leopard, elephant, Great Indian Bustard, Indian rhinoceros.
- **Schedule II** — protected species with lower penalties.
- **Schedule III** — protected plants.
- **Schedule IV** — CITES-listed species (invasive alien species also included after 2022).

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**WILDLIFE PROTECTION
ACT 1972 — SCHEDULES**

Schedule I

HIGHEST protection: no
hunting at all
Tiger, Elephant, Lion,
Rhino, Snow Leopard
Gharial, Gangetic Dolphin,
Golden Langur, Great
Indian Bustard

Punishment: minimum 3
years; up to 7 years
Fine up to Rs 25,000
(amended amounts vary)
Repeat offenders:
minimum 7 years

Schedules II and III

Schedule II: high
protection; lesser penalty
than I
Schedule III: specified
animals; hunting allowed
with license
Schedule IV: lesser
protection; license required

Schedules V and VI

Schedule V: vermin (can
be hunted)
Common crow, fruit bat,
common rat, mice
Schedule VI: protected
plants (6 species originally)

2022 Amendment

WPA Amendment 2022
consolidated to 4 schedules
Also incorporated CITES
(Convention on Int'l Trade
in Endangered Species)
into Indian law

Chapter F3 — Key Institutions

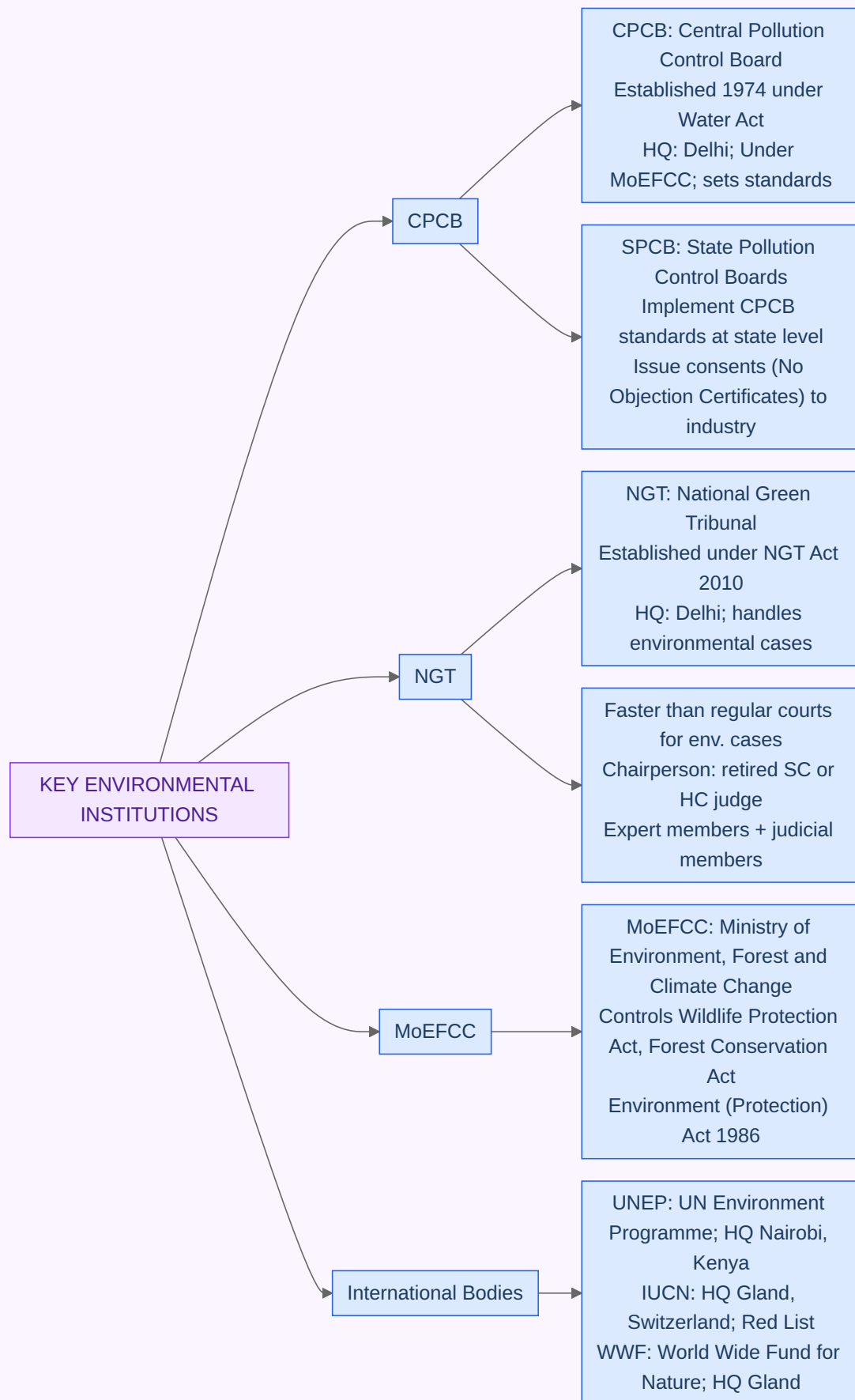
Institution	Year	Role
CPCB	1974	Central Pollution Control Board; monitors and controls air + water pollution
MoEFCC	—	Ministry of Environment, Forest & Climate Change — apex authority
NBA	2003	National Biodiversity Authority; HQ Chennai
NTCA	2005	National Tiger Conservation Authority; Project Tiger administration
WII	1982	Wildlife Institute of India; HQ Dehradun
ZSI	1916	Zoological Survey of India; HQ Kolkata
BSI	1890	Botanical Survey of India; HQ Kolkata
FSI	1981	Forest Survey of India; publishes India State of Forest Report (biennial); HQ Dehradun
NGT	2010	National Green Tribunal; 5 benches (Delhi principal + 4 regional)
NMCG	2011	National Mission for Clean Ganga; implements Namami Gange Programme
WCCB	2007	Wildlife Crime Control Bureau — enforces wildlife laws, busts poaching networks

India State of Forest Report (ISFR) — Key Data

- Published by FSI **every 2 years**.
- Total forest + tree cover = **25.17% of geographical area** (target under National Forest Policy 1988 = 33%).
- Forest cover alone = 21.76% (~7.16 lakh km²).
- Largest forest cover (area): MP > Arunachal > Chhattisgarh > Odisha > Maharashtra.**
- Highest forest cover (% of state area): Mizoram > Lakshadweep > A&N > Meghalaya > Arunachal.**
- Mangrove cover: ~4,992 km². Gujarat > West Bengal > A&N (by area).

PART X — CONSERVATION PROJECTS AND NATIONAL SYMBOLS

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Conservation Projects

Project	Year	Focus	Current count
Project Tiger	1973	Tiger conservation; NTCA manages	58 Tiger Reserves (April 2026)
Crocodile Conservation	1975	Gharial, mugger, saltwater crocodile	—
Project Elephant	1992	Asian elephant conservation	33 Elephant Reserves
Project Snow Leopard	2009	Cold-desert ecosystem; snow leopard	—
Project Lion	2020	Asiatic lion (only at Gir, Gujarat)	—
Project Dolphin	2020	Gangetic river dolphin + marine dolphins	—
Project Cheetah	2022	World's first intercontinental cheetah translocation; Kuno NP (MP)	8 (Namibia) + 12 (South Africa)
Vulture Conservation	2006	Breeding centres; phase-out of diclofenac (the drug that decimated vultures)	—

KEY FACT

Project Tiger (1973) — India's oldest and most famous conservation project. First reserve was **Bandipur (Karnataka)**. Madhya Pradesh has the most tiger reserves (9) and the most tigers (~785 as per 2022 census). Project Cheetah (2022) is the world's first intercontinental translocation of a large carnivore — cheetahs were extinct in India since 1952.

National Symbols of India

Category	Symbol
National animal	Royal Bengal Tiger (<i>Panthera tigris tigris</i>)
National bird	Indian Peafowl (<i>Pavo cristatus</i>)
National aquatic animal	Gangetic River Dolphin (<i>Platanista gangetica</i>) — declared 2009
National heritage animal	Indian Elephant (<i>Elephas maximus indicus</i>) — declared 2010
National flower	Lotus (<i>Nelumbo nucifera</i>)
National tree	Indian Banyan (<i>Ficus benghalensis</i>)
National fruit	Mango (<i>Mangifera indica</i>)
National river	Ganga — declared 2008
National reptile	King Cobra (<i>Ophiophagus hannah</i>)
National butterfly	Southern Birdwing (<i>Troides minos</i>)

PART Y — INTERNATIONAL CONVENTIONS (COMPLETE TABLE)

Year	Convention	Key outcome
1971	Ramsar Convention	Conservation of wetlands; India joined 1982; 85+ Indian sites
1972	Stockholm Conference	First UN env conference; UNEP formed (HQ Nairobi); World Environment Day = 5 June
1973	CITES	Trade in endangered species; 3 appendices; India joined 1976
1979	Bonn Convention (CMS)	Conservation of migratory species; India 1983
1985	Vienna Convention	Framework for ozone layer protection
1987	Montreal Protocol	Phase-out CFCs and other ODSs; India 1992; most successful env treaty
1989	Basel Convention	Trans-boundary movement of hazardous wastes; India 1992
1992	Rio Earth Summit (UNCED)	Agenda 21; UNFCCC, CBD signed; birth of modern env diplomacy
1992	CBD	Convention on Biological Diversity; conservation + sustainable use + benefit-sharing
1992	UNFCCC	UN Framework Convention on Climate Change; entered force 1994
1997	Kyoto Protocol	Binding GHG cuts on Annex-I (developed) countries only
2000	Cartagena Protocol	Biosafety of living modified organisms (LMOs); under CBD
2001	Stockholm Convention (POPs)	"Dirty dozen" persistent organic pollutants; India 2006
2010	Nagoya Protocol	Access + benefit sharing of genetic resources; under CBD
2010	Aichi Biodiversity Targets	20 targets under CBD for 2020
2013	Minamata Convention	Reduce mercury use and emissions; India 2018
2015	Paris Agreement	Limit warming to 1.5–2°C; voluntary NDCs by all parties
2015	SDGs	17 Sustainable Development Goals; 169 targets; UN GA
2016	Kigali Amendment	Phase-down HFCs (added to Montreal Protocol); India 2021
2021	Glasgow COP-26	India's PANCHAMRIT; coal phase-down
2022	Sharm el-Sheikh COP-27	Loss and Damage fund agreed in principle
2022	Kunming-Montreal GBF	"30×30" — 30% of land + sea protected by 2030; CBD COP-15
2023	Dubai COP-28	First Global Stocktake; "transition away from fossil fuels"
2024	Baku COP-29	\$300 billion/year climate finance by 2035

Year	Convention	Key outcome
2025	Belém COP-30	Brazil host; first COP in the Amazon

PART Z — BIODIVERSITY (DETAILED)

Famous Environmental Movements

Movement	Year	Location	Leader(s)
Chipko movement	1973	Garhwal, Uttarakhand	Sundarlal Bahuguna, Chandi Prasad Bhatt
Silent Valley movement	1973–83	Kerala	Saved Silent Valley from a dam
Appiko movement	1983	Karnataka (Western Ghats)	Pandurang Hegde
Narmada Bachao Andolan	1985–	MP / Maharashtra / Gujarat	Medha Patkar , Baba Amte
Tehri Dam protest	1990s	Uttarakhand	Sundarlal Bahuguna

Global Environmental Days

Date	Day
2 February	World Wetlands Day
3 March	World Wildlife Day
21 March	International Day of Forests
22 March	World Water Day
22 April	Earth Day
22 May	International Day for Biological Diversity
5 June	World Environment Day
8 June	World Oceans Day
16 September	World Ozone Day
13 October	International Day for Disaster Risk Reduction

Key Acronyms

Acronym	Full Form
UNFCCC	UN Framework Convention on Climate Change
IPCC	Intergovernmental Panel on Climate Change
UNEP	UN Environment Programme
CBD	Convention on Biological Diversity
CITES	Convention on International Trade in Endangered Species
GEF	Global Environment Facility
IUCN	International Union for Conservation of Nature
COP	Conference of the Parties
NDC	Nationally Determined Contribution
REDD+	Reducing Emissions from Deforestation and forest Degradation
CPCB	Central Pollution Control Board
MoEFCC	Ministry of Environment, Forest & Climate Change
NGT	National Green Tribunal
NBA	National Biodiversity Authority
NTCA	National Tiger Conservation Authority
FSI	Forest Survey of India

Acronym	Full Form
WII	Wildlife Institute of India
CBDR	Common But Differentiated Responsibilities
GBF	Global Biodiversity Framework (Kunming-Montreal 2022)

APPENDIX A — Disaster Management (Brief)

- **Disaster Management Act 2005** — legal framework for disaster management in India.
- **NDMA** (National Disaster Management Authority) — chaired by the Prime Minister.
- **NDRF** (National Disaster Response Force) — operational since 2006; 16 battalions.
- **CDRI** (Coalition for Disaster Resilient Infrastructure) — launched by India in 2019; international initiative.

India's seismic zones: **Zones II to V** (4 zones by BIS). Zone V is the highest risk — covers J&K (Kashmir valley), parts of HP, Uttarakhand, all northeastern states, and Kutch region of Gujarat.

APPENDIX B — SDGs (Sustainable Development Goals)

17 Goals adopted by UN General Assembly in 2015, with 169 targets, to be achieved by 2030:

1. No Poverty | 2. Zero Hunger | 3. Good Health | 4. Quality Education | 5. Gender Equality | 6. Clean Water | 7. Clean Energy | 8. Decent Work | 9. Industry & Innovation | 10. Reduced Inequalities | 11. Sustainable Cities | 12. Responsible Consumption | 13. Climate Action | 14. Life Below Water | 15. Life on Land | 16. Peace & Justice | 17. Partnerships

SDG India Index is published annually by NITI Aayog.

APPENDIX C — 25-Question Mini-Mock

1. What year was the Stockholm Conference on Human Environment held?
 2. Which city hosts UNEP headquarters?
 3. When is World Environment Day celebrated?
 4. What is the Ramsar Convention about and in what year was it signed?
 5. India's first Ramsar site(s)?
 6. How many biodiversity hotspots does India have?
 7. When was the Bhopal gas tragedy and what chemical leaked?
 8. What year was India's Environment Protection Act passed?
 9. What year was the Wildlife Protection Act passed?
 10. How many schedules does the WLPA now have (after 2022)?
 11. When was NTCA established?
 12. What year was Project Tiger launched?
 13. What is India's national aquatic animal?
 14. What is India's national heritage animal?
 15. What year was the Montreal Protocol signed?
 16. What year was the Kyoto Protocol signed?
 17. What year was the Paris Agreement signed?
 18. What is India's net-zero target year?
-

19. What is the "30×30" target and which agreement does it come from?

20. What does PANCHAMRIT refer to?

21. Which Act created the National Green Tribunal?

22. What is India's current forest cover as a percentage of its area?

23. How many SDGs are there and by when must they be achieved?

24. India's largest biosphere reserve by area is in which state?

25. Name the only floating National Park in India and its state.

Answers: 1. 1972 2. Nairobi, Kenya 3. 5 June 4. Conservation of wetlands; 1971 (at Ramsar, Iran) 5. Chilika (Odisha) and Keoladeo Ghana (Rajasthan) — both designated 1981 6. 4 (Himalayas, Indo-Burma, Western Ghats+Sri Lanka, Sundaland) 7. 2–3 December 1984; Methyl Isocyanate (MIC) from Union Carbide 8. 1986 9. 1972 10. 4 schedules 11. 2005 12. 1973 13. Gangetic River Dolphin (declared 2009) 14. Indian Elephant (declared 2010) 15. 1987 16. 1997 17. 2015 18. 2070 (announced at COP-26, Glasgow, 2021) 19. Protecting 30% of Earth's land and sea by 2030; Kunming-Montreal Global Biodiversity Framework (2022) 20. India's 5 climate commitments made at Glasgow COP-26 (2021) 21. National Green Tribunal Act, 2010 22. 25.17% (forest + tree cover) 23. 17 goals; by 2030 24. Gujarat (Kachchh Biosphere Reserve is the largest) 25. Keibul Lamjao National Park; Manipur (on Loktak Lake)

APPENDIX D — Trap Recognition Card

Trap	Common mistake	Correct answer
EPA year	1984 (Bhopal year)	1986 — passed 2 years after Bhopal
Stockholm — which one?	Mixing 1972 conf and 2001 POPs convention	1972 = Conference on Human Env; 2001 = POPs Convention
Vienna vs Montreal	Same thing	Different: Vienna 1985 = framework; Montreal 1987 = actual CFC phase-out
Kyoto binds whom?	All countries	Only Annex-I (developed) countries
India's first Ramsar site	Chilika only	Both Chilika AND Keoladeo (1981)
WLPAs schedules	6	4 (after 2022 amendment)
India joined Montreal Protocol	1987	1992 — India joined 5 years after the protocol was signed
India's net-zero target	2050 or 2060	2070
First NP of India	Jim Corbett established directly	Hailey NP (1936) → renamed Jim Corbett
Only Asiatic lions	Multiple locations	Gir NP, Gujarat — the ONLY place in the world
Project Tiger first reserve	Corbett	Bandipur, Karnataka (1973)
Biosphere Reserves on UNESCO list	All 18	Only 12 of India's 18 BRs are on UNESCO MAB World Network
Floating NP	Some river-based NP	Keibul Lamjao (Manipur, Loktak Lake) is the ONLY one
Largest NP in India	Sundarbans or Kaziranga	Hemis NP (Ladakh, ~4,400 km ²)